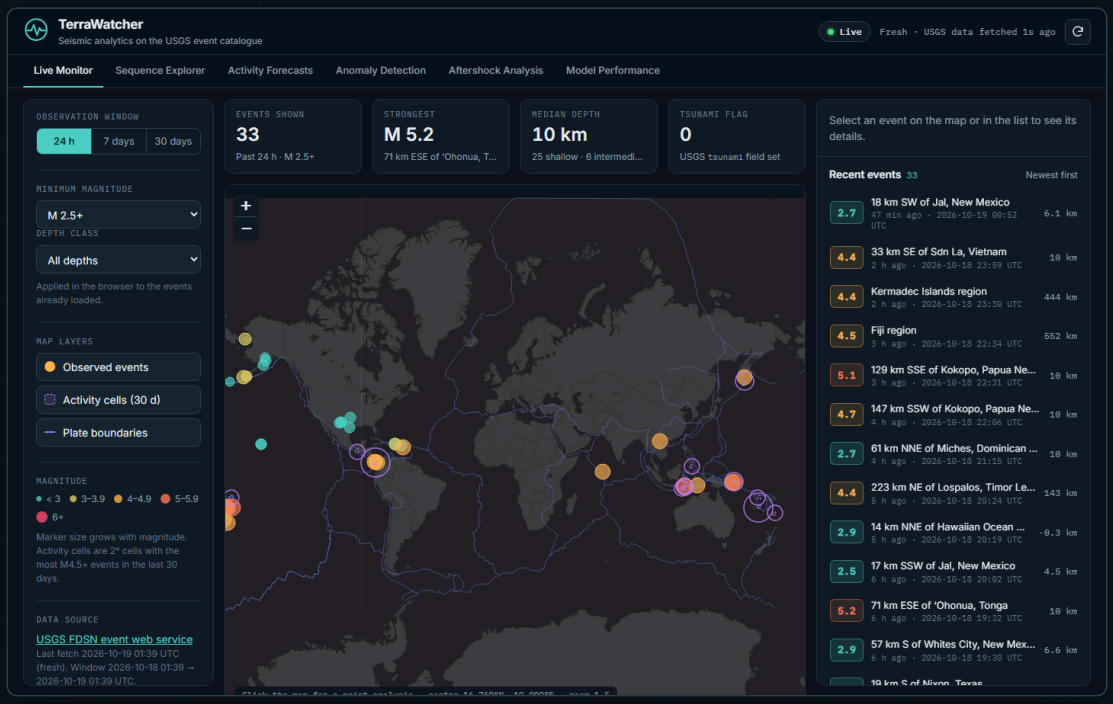
TECHNICAL LEARNING GUIDE

TerraWatcher

Seismic analytics on the USGS earthquake catalogue



Architecture · Data pipeline · Geospatial concepts · Sequences · Forecasts · Anomalies · Aftershocks · Scientific validity · Deployment · Testing · Interview preparation

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Written against the repository state of 19 October 2026 · Flask · Leaflet · scikit-learn · SciPy · PyTorch · USGS FDSN

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1 Project overview

1.1 What TerraWatcher does

TerraWatcher is a web application. It downloads earthquake records from the United States Geological Survey (USGS), shows them on an interactive world map and summarises them with live statistics. On top of that live layer it runs four analytical modules: sequence detection, seven-day regional count forecasts, elevated-activity detection and aftershock (Omori–Utsu) analysis. Each module is evaluated on data it was not built from, and its results are shown in a Model Performance view (Chapter 8). A small legacy classifier from the original project is kept for the point-analysis dialog (Chapter 7).

In one sentence for an interviewer: *TerraWatcher is a Flask and Leaflet seismic-analytics dashboard on live USGS data, with space-time clustering, walk-forward-evaluated count forecasts against baselines, a multiple-comparison-corrected anomaly test and maximum-likelihood aftershock fits, each with honest failure states.*

The application produces four kinds of output, and keeping them apart is the most important idea in this guide:

Layer	Example in TerraWatcher	Source of truth	Can it be wrong?
Observed data	"An M 5.2 earthquake occurred 71 km ESE of 'Ohonua, Tonga at 19:32 UTC"	USGS catalogue, passed through unchanged	Only if USGS revises it (records have a `status` of `automatic` or `reviewed`)
Derived statistics and patterns	median depth, magnitude histogram, "63 M4.5+ events in this 2° cell in 30 days"	Computed by TerraWatcher from observed data	Only through bugs; the arithmetic is deterministic
Statistical findings	"This cell recorded 67 events against ~4.2 expected ($q = 2.4 \times 10^{-10}$)"; "these 86 events form one detected sequence"	Anomaly test and DBSCAN on observed data	Only through bugs or wrong assumptions; the assumptions are stated
Model estimates	"11.3 events expected in this cell over the next 7 days (80 %: 1–26)"; "If an M4.5+ event were recorded here, 28 % chance it is M5+"	Forecast models and `EarthquakeNN`, trained on past data	Yes, often. They are statistical estimates with measured, limited skill

Nothing in the application predicts that an individual earthquake will happen at a particular place or time. The forecasts are aggregate weekly counts for 10° cells.

1.2 The problem it addresses

Raw seismic catalogues are hard for non-specialists to read. The USGS FDSN event service returns GeoJSON with dozens of fields per event, epoch-millisecond timestamps and `[longitude, latitude, depth]` arrays. TerraWatcher turns that feed into something a person can explore in seconds: where earthquakes are happening now, how deep they are, which areas have been unusually active this month, and how those events relate to plate boundaries.

As an engineering project, it demonstrates several skills:

- **Integration.** It consumes an external API robustly, with timeouts, retries, validation, de-duplication and stale-data handling.
- **Geospatial computation.** It handles coordinate order, antimeridian wrap-around, approximate distances on a sphere and spatial binning.
- **Honest ML.** It defines a label carefully, avoids leakage, splits data chronologically, compares against baselines and reports metrics that are not flattering.
- **Frontend work.** It manages state, cancels superseded requests, and has accessible controls and clear loading, empty and error states.

1.3 Intended users and use cases

- **Students and learners** exploring global seismicity and plate tectonics.

- **Recruiters and interviewers** evaluating the project as a portfolio piece; the demo checklist in Chapter 16 is written for them.
- **The developer** (you), who needs to explain, debug and extend the system.

TerraWatcher is *not* intended for emergency decisions. The footer and every model output say so.

1.4 Implemented features

Feature	Where it lives	Kind of output
Event feed for the last 24 h / 7 d / 30 d with a minimum-magnitude filter	`/api/earthquakes`, `loadFeed()`	Observed
Map markers sized and coloured by magnitude, with tooltips	`renderMarkers()`	Observed
Event list (newest first, up to 200 rows) and an event detail panel	`renderList()`, `renderDetail()`	Observed
Client-side depth-class filter (shallow / intermediate / deep)	`core.filterByDepth()`	Derived
Metrics: count, strongest event, median depth, tsunami-flag count	`core.summarize()`	Derived
Magnitude and depth histograms	`renderBars()`	Derived
Activity clusters: 2° cells with the most M4.5+ events in 30 days	`/api/hotspots`	Derived (historical pattern)
Plate-boundary overlay	`/api/plates`, `loadPlates()`	Reference data
Point analysis: conditional severity score at any coordinate	`/api/predict`, `scorePoint()`	Model estimate
Legacy classifier metrics (now in Model Performance)	`/api/model-info`, `/api/model-performance`	Model evaluation
Freshness labels (fresh / cached / stale), truncation notice, error banner with Retry	`renderFreshness()`, `loadFeed()`	System status
Health endpoint with per-module readiness	`/api/health`	System status
Sequence Explorer: DBSCAN sequences, statistics, members, parameters	`/api/sequences`, `ml.js`	Statistical finding
Activity Forecasts: 7-day counts per 10° cell, intervals, baselines, backtest	`/api/forecast[/<cell>]`	Model estimate
Anomaly Detection: elevated observed activity, p/q values, contributing events	`/api/anomalies[/<cell>]`	Statistical finding
Aftershock Analysis: timeline, histograms, Omori–Utsu fit with diagnostics, comparison	`/api/aftershocks`	Statistics + model
Model Performance: validation vs test metrics of every module	`/api/model-performance`	Model evaluation

1.5 Technology stack and why each piece is used

Technology	Role	Why this choice
Python 3.12	Backend language	Mature scientific ecosystem; the project's venv uses 3.12.10.
Flask 3.1	HTTP server and routing	Small and explicit. About a dozen JSON routes and one HTML page do not need a larger framework.
requests	HTTP client for USGS	Simple API with timeouts; easy to replace with a fake in tests.
NumPy / pandas	Vectorised feature computation, training data frames	Plate distance for thousands of points runs as array arithmetic instead of Python loops.
PyTorch	The neural network	Already used by the original project; small models train in seconds on a CPU.
scikit-learn	DBSCAN, Poisson GLM, Poisson gradient boosting, `StandardScaler`, metrics	The standard, well-tested implementations.
SciPy	Negative-binomial and Poisson distributions, Omori–Utsu MLE (L-BFGS-B), KS and binomial tests	Already required by scikit-learn; now pinned explicitly because it is used directly.
Leaflet 1.9.4	Interactive map	Lightweight, no API key, canvas rendering for thousands of markers.
Plain HTML/CSS/JS	Frontend	No build step. Pure logic is isolated in `core.js` so it can be unit-tested with Node.
pytest / node:test	Tests	pytest for Python; Node's built-in runner, so the frontend needs no npm dependencies.

Deliberately *absent*: a database, Redis, background workers, Docker and a frontend framework. Each would add operational weight without solving a problem the project has (see Chapter 12).

1.6 Monitoring, analysis, hotspots and forecasting

These four words are often blurred together. TerraWatcher keeps them separate:

1. **Monitoring.** Showing what has been recorded. TerraWatcher does this with real USGS data. Its freshness is limited by USGS processing delays (minutes for automatic solutions) plus the 60-second feed cache.
2. **Historical analysis.** Describing past events, for example "most events this week were shallow". TerraWatcher does this for the loaded window only (at most 30 days).
3. **Hotspot and anomaly analysis.** TerraWatcher's "activity cells" are a *count of recent observed events per grid cell*. The anomaly module goes one step further and tests whether a cell's count is unusual for that cell. Both describe where activity *has been*, not where it *will be*.
4. **Forecasting.** The forecast module issues seven-day count forecasts for 10° cells and was evaluated walk-forward against baselines on 2020–2026 (§8.4). That is a statistical forecast of aggregate activity, with measured and modest skill. It is not a prediction of individual earthquakes, and it is not a prospective test in the CSEP sense (Chapter 9 describes what that would involve).

1.7 What the application can and cannot claim

Can claim:

- Its M4.5 weekly count forecasts had 13.9 % lower mean Poisson deviance than the historical-mean baseline on the 2020–2026 walk-forward test, and were better in every test year, but not in every cell (§8.4).
- Its sequence detector recovers known synthetic clusters, including across the antimeridian and near the poles (§8.3). Its anomaly test keeps run-level false alerts near the nominal 5 % (5.6 % in simulation), at the cost of low sensitivity (§8.5).
- It displays USGS events with their reported coordinates, magnitudes, depths and times.
- It reports when the data was fetched and whether a response came from cache or is stale.

- Its model, *for events that have already been recorded at M ≥ 4.5*, separates M ≥ 5 events from smaller ones slightly better than chance. On the 2026-07-11 → 2026-10-18 test period it scored ROC-AUC 0.570, against 0.552 for logistic regression and 0.500 for a constant base rate.

Cannot claim:

- That any location is "at risk" of an earthquake on any time scale, or that a forecast or an elevated flag anticipates a large earthquake.
- That the model has practical predictive value. Its Brier skill over the base rate is +0.006, which is close to zero.
- That empty areas of the map are safe. An empty map only means USGS recorded nothing above the chosen magnitude in the chosen window.

1.8 High-level architecture

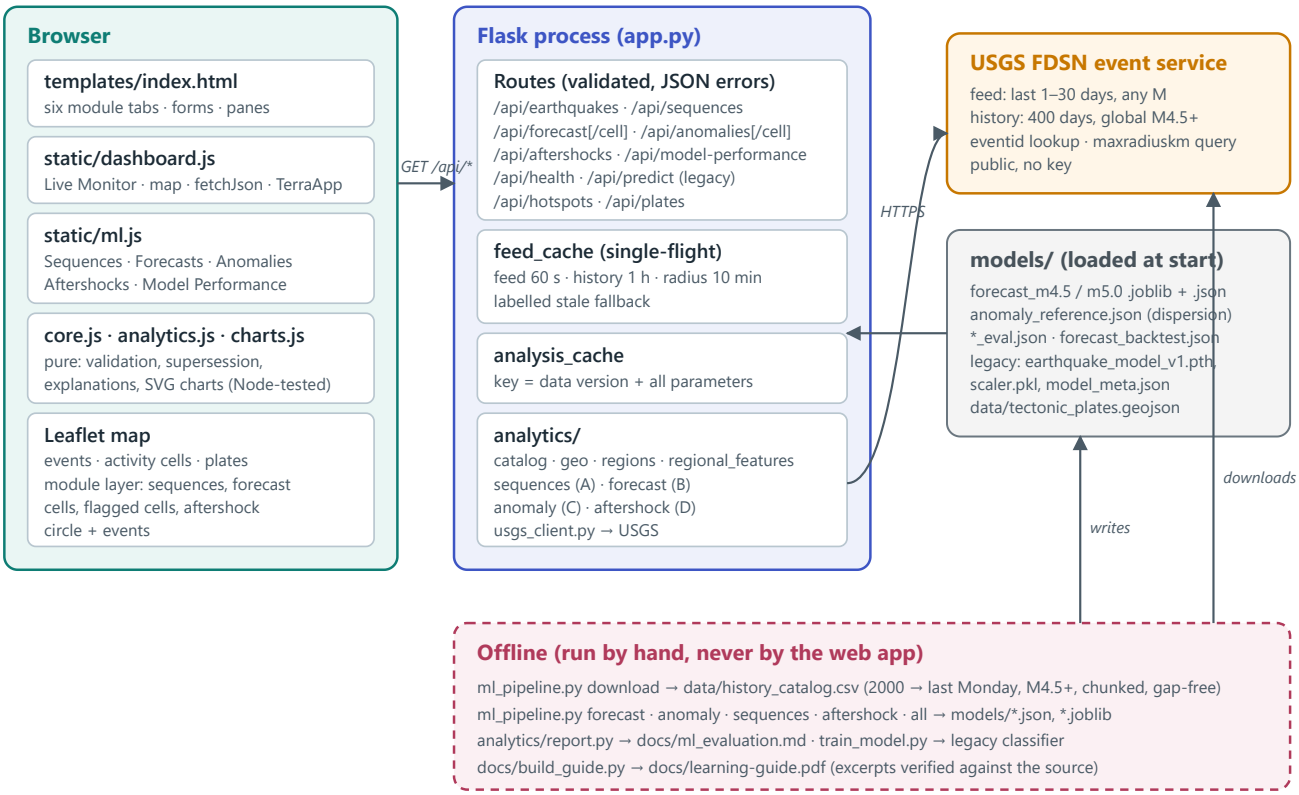


Figure 1. Components that actually exist. The browser talks only to Flask (same origin), except for map tiles and the Leaflet library. Flask talks to USGS through `usgs_client.py`. Models, evaluation files and plate data are loaded once at start-up. Training and evaluation are offline commands run by hand.

There is **one process** at runtime: `python app.py`. It serves the HTML page, the static JavaScript and CSS, and the JSON API. State lives in memory: the two caches, the loaded models and evaluation files, and the plate segments. Restarting the process clears the caches and reloads the artifacts.

2 Repository structure

2.1 Directory map

```
.
├── app.py                    Flask application: routes, validation, cache, errors, headers
├── config.py                 Configuration constants + environment-variable overrides
├── usgs_client.py            USGS FDSN client: HTTP, retry, validation, normalisation, event lookup, radius search
├── ml_pipeline.py            Offline CLI for the analytical modules (download/forecast/anomaly/sequences/aftershock/all)
├── analytics/                The four analytical modules and their shared data layer (Chapter 8)
│   ├── catalog.py, geo.py, regions.py, regional_features.py
│   ├── sequences.py, forecast.py, forecast_training.py, anomaly.py, aftershock.py
│   └── evaluation.py, report.py
├── models/                   Forecast models, anomaly reference, evaluation results (generated, committed)
├── ruff.toml                 Lint configuration (development only)
├── features.py               Legacy classifier features, shared by training and serving
├── ml_model.py               EarthquakeNN + ModelBundle (artifact loading, inference)
├── train_model.py            Offline training and evaluation pipeline
├── earthquake_model_v1.pth   Trained PyTorch weights (state_dict)
├── scaler.pkl                Fitted sklearn StandardScaler
├── model_meta.json           Feature schema, label definition, split, metrics, baselines
├── requirements.txt          Runtime dependencies
├── requirements-dev.txt      + pytest
├── requirements-docs.txt     Tools for building this guide
├── pytest.ini                Test discovery + warning policy
├── data/
│   └── tectonic_plates.geojson   Plate boundaries (241 features, 6051 segments)
├── templates/
│   └── index.html            Dashboard markup (Jinja template, no server-side logic)
├── static/
│   ├── core.js               Pure browser logic (validation, stats, formatting)
│   ├── charts.js             Dependency-free SVG charts (pure, Node-tested)
│   ├── analytics.js          Analytical-view helpers (pure, Node-tested)
│   ├── ml.js                 The five analytical views
│   ├── dashboard.js          Live Monitor: state, networking, Leaflet layers, rendering
│   └── dashboard.css         Theme and responsive layout
├── tests/                    pytest suites + tests/frontend/*.test.js
└── docs/                     This guide, ml_evaluation.md, model evaluation, audit, diagrams, screenshots
```

Not documented here because they are generated or environment-specific: `.venv/`, `__pycache__/`, `.pytest_cache/`, `docs/build/` (PDF build intermediates), `data/training_catalog.csv` (the legacy classifier's catalogue, recreated by `train_model.py`) and `data/history_catalog.csv` + `.json` (the 2000–2026 catalogue, recreated by `python ml_pipeline.py download`). All of these are git-ignored.

2.2 File-by-file reference

File	Responsibility	Inputs → outputs	Depends on	Used by
`config.py`	Paths, timeouts, TTLs, limits	Environment variables → constants	`os`	every Python module
`usgs_client.py`	Talk to USGS safely	`start`, `end`, `min_mag`, `limit` → `(events, stats)` or `UpstreamError`	`requests`, `config`	`app.py`, `train_model.py`
`features.py`	Define the five features	Coordinates, times, catalogue → arrays	`numpy`	`app.py`, `train_model.py`, `ml_model.py` (names only)
`ml_model.py`	Network class; load and validate artifacts; predict	Artifact paths → `ModelBundle`; feature matrix → probabilities	`torch`, `joblib`, `features`	`app.py`, `train_model.py`
`app.py`	HTTP API and page	Requests → JSON/HTML	all of the above, `flask`	browser; tests
`train_model.py`	Build dataset, train, evaluate, save	USGS (or CSV) → `.pth`, `.pkl`, `model_meta.json`, `docs/model_evaluation.md`	`usgs_client`, `features`, `ml_model`, `sklearn`, `pandas`	run manually
`templates/index.html`	Page structure, accessible controls	— → DOM	Leaflet CDN, `core.js`, `dashboard.js`	Flask `/` route
`static/core.js`	Pure functions	API payloads → validated events, statistics, strings	none	`dashboard.js`, Node tests
`static/dashboard.js`	Behaviour	User events, API JSON → DOM, Leaflet layers	`core.js`, Leaflet	browser
`static/dashboard.css`	Presentation	—	—	browser
`data/tectonic_plates.geojson`	Reference geometry	—	—	`features.load_plate_segments`, `/api/plates`
`model_meta.json`	Contract between training and serving (legacy classifier)	written by training	—	`ModelBundle`, `/api/model-info`, Model Performance view
`tests/conftest.py`	Fake USGS transport, fixtures	—	`usgs_client`, `app`	all API tests
`analytics/*.py`	The analytical modules (Chapter 8)	events, count matrices → features, sequences, forecasts, tests, fits	`numpy`, `pandas`, `scipy`, `sklearn`	`app.py`, `ml_pipeline.py`
`ml_pipeline.py`	Offline training and evaluation	`data/history_catalog.csv` → `models/*`, `docs/ml_evaluation.md`	`analytics`	run manually
`static/ml.js`	The analytical views	API JSON → DOM, module map layer	`analytics.js`, `charts.js`,	browser

File	Responsibility	Inputs → outputs	Depends on	Used by
			`dashboard.js`	

2.3 How a request moves through the files

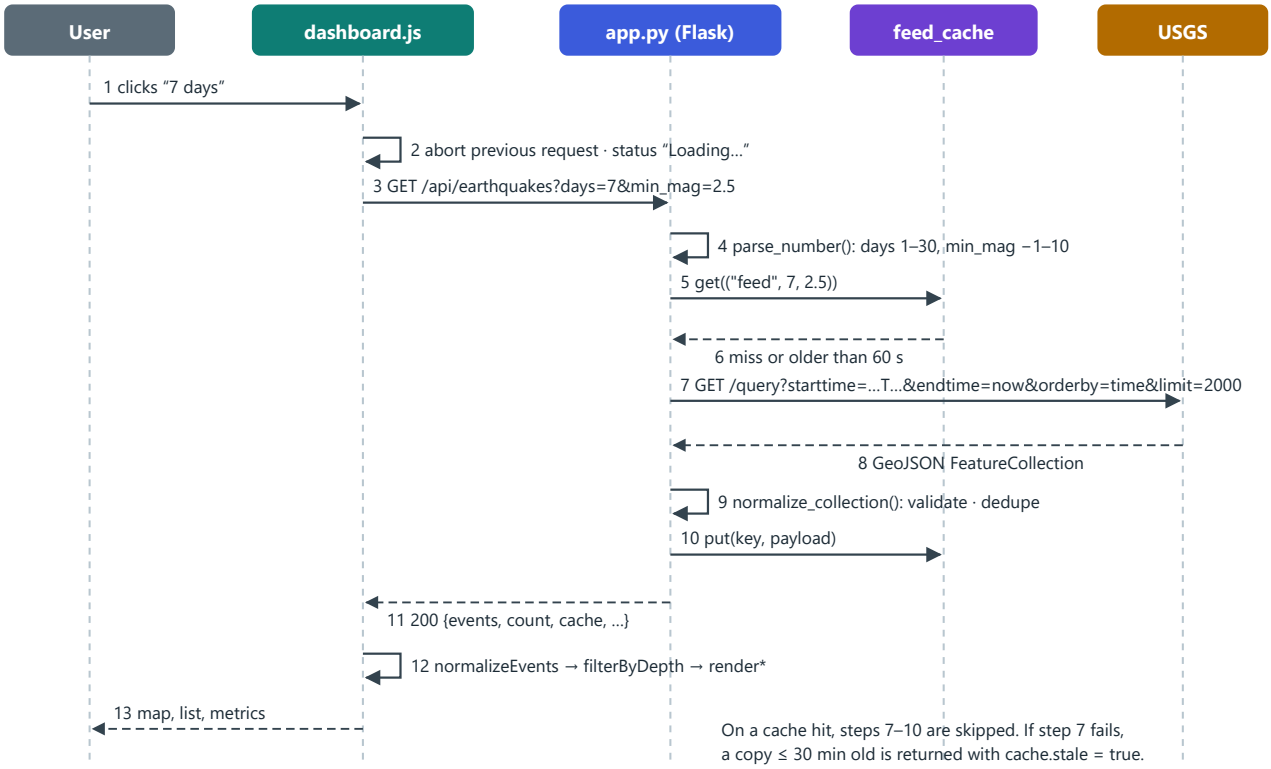


Figure 2. What happens when the user switches the observation window to 7 days. The numbered steps correspond to real functions: `loadFeed()` (1–3), `parse_number()` (4), `cached_fetch()` (5–6, 10), `usgs_client.fetch_events()` (7–9), `normalizeEvents()` and the render functions (12).

Two other flows reuse the same pieces:

- **Point analysis** (`/api/predict`): `parse_number` validates the input, `get_context_catalog()` uses `cached_fetch` with a 10-minute TTL, `score_points()` calls `features.py` , and `ModelBundle.predict_proba` produces the probability.
- **Activity clusters** (`/api/hotspots`): the same context catalogue is binned into 2° cells in `hotspots()` , and the busiest cells are scored with `score_points()` .
- **Forecasts and anomalies** (`/api/forecast` , `/api/anomalies`): `get_history()` downloads 400 days of global M4.5+ events through `cached_fetch` (1-hour TTL, single flight). The weekly count matrix is built from it, and the result is memoised under `(module, parameters, fetched_at)` .
- **Sequences** (`/api/sequences`): `get_feed()` (the same cached feed as `/api/earthquakes`) is passed to `analytics.sequences.detect` .
- **Aftershocks** (`/api/aftershocks`): `usgs_client.fetch_event` looks up the reference event, a `circle=` query fetches the surroundings, and `analytics.aftershock.analyse` computes the statistics and the fit.

3 Backend architecture

3.1 Entry point and start-up sequence

`python app.py` executes the module from top to bottom:

1. `logging.basicConfig(...)` configures log output (level from `TERRAWATCHER_LOG_LEVEL`).
2. `app = Flask(__name__)`; `app.json.sort_keys = False` keeps the JSON key order readable.
3. The error handlers and the `after_request` security-header hook are registered by decorators.
4. `plate_segments = F.load_plate_segments(config.TECTONIC_FILE)` reads the 6051 boundary segments.
5. `bundle = ModelBundle(...)` loads and validates the three model artifacts. If anything is wrong, the app **still starts**. The model is marked unavailable, and the routes that need it return 503.
6. `app.run(host=config.HOST, port=config.PORT, debug=config.DEBUG, threaded=True)` starts Werkzeug's development server, with one thread per request.

app.py

```
plate_segments = F.load_plate_segments(config.TECTONIC_FILE)
bundle = ModelBundle(config.MODEL_FILE, config.SCALER_FILE, config.MODEL_META_FILE)
...
if __name__ == "__main__":
    app.run(host=config.HOST, port=config.PORT, debug=config.DEBUG, threaded=True)
```

Nothing is trained or downloaded at start-up. The first USGS call happens when the first browser request arrives, so start-up takes about two seconds and does not depend on the network.

> NOTE Because `plate_segments` and `bundle` are module-level globals, every request thread shares them. Both are read-only after start-up, so sharing them is safe. The only mutable shared state is `feed_cache`, which is protected by a lock (§3.6).

3.2 Configuration

All tunables live in `config.py`. Each one can be overridden with an environment variable, and malformed values fall back to the default instead of crashing:

config.py

```
def _env_float(name, default):
    try:
        return float(os.environ.get(name, default))
    except (TypeError, ValueError):
        return float(default)
...
USGS_TIMEOUT = _env_float("TERRAWATCHER_USGS_TIMEOUT", 15)   # seconds per request
USGS_RETRIES = _env_int("TERRAWATCHER_USGS_RETRIES", 1)      # retries on 429/5xx/timeouts
```

The defaults bind to `127.0.0.1` (not reachable from other machines), keep debug mode off and require no secrets. CORS is not configured because the frontend is served by the same Flask process: the browser treats `/api/*` as same-origin, so no `Access-Control-Allow-Origin` header is needed. A test asserts that none is sent.

3.3 Endpoint reference

All endpoints accept only `GET`. Every error, from any endpoint, has the same JSON shape:

```
{"error": "Parameter 'lat' must be between -90.0 and 90.0", "code": "out_of_range", "status": 400}
```

The frontend shows `error` to the user. Programs and tests can branch on `code`.

The analytical endpoints are summarised in §16.3 and their parameters in the README. Their error codes follow one pattern:

- invalid or out-of-range parameters give 400 (`invalid_parameter` / `out_of_range`);
- malformed region ids give 404 `unknown_region` ;
- cells without enough history give 404 `insufficient_history` or `no_activity` ;
- an unknown USGS event gives 404 `event_not_found` ;
- missing models give 503 `model_unavailable` ;
- a failed or truncated history download gives 503 `history_unavailable` , with the text “No result was computed”.

GET /api/earthquakes

Item	Value
Purpose	Observed events in `[now − days, now)` , newest first
Parameters	`days` integer 1–30 (default 1); `min_mag` number −1 to 10 (default 2.5, rounded to 0.1)
Success	200: `events[]` , `count` , `truncated` , `total_available` , `query{days,min_mag,start,end,limit}` , `fetched_at` , `dropped_invalid` , `duplicates_removed` , `source` , `cache{hit,age_seconds,stale,warning}`
Errors	400 `invalid_parameter` / `out_of_range` ; 502 `upstream_*` (unreachable, 5xx, rate limited, malformed); 504 `upstream_timeout`
Source	`app.py` → `earthquakes()`

Internally the handler validates both parameters and builds a cache key `("feed", days, min_mag)` . It then calls `cached_fetch` , which either returns a cached payload younger than 60 s or calls `fetch()` . On a miss, `fetch()` requests up to 2000 events from USGS. If exactly 2000 come back, it asks the USGS `count` endpoint how many exist in total, so the UI can say “newest 2000 of 9264”.

Real request on 2026-10-18, first event shown:


```
{"events": [{"id": "us6000u37h", "time": 1792351948734, "time_iso": "2026-10-18T19:32:28.734Z",
   "latitude": -21.541, "longitude": -174.2956, "depth_km": 10.0, "mag": 5.2, "mag_type": "mb",
   "place": "71 km ESE of ʻOhonua, Tonga", "status": "reviewed", "tsunami": false, "...": "..."}],
 "count": 7, "truncated": false, "total_available": 7,
 "query": {"days": 1, "min_mag": 4.5, "start": "2026-10-17T21:38:24Z",
           "end": "2026-10-18T21:38:24Z", "limit": 2000},
 "fetched_at": "2026-10-18T21:38:24Z", "dropped_invalid": 0, "duplicates_removed": 0,
 "source": "USGS FDSN event web service",
 "cache": {"hit": false, "age_seconds": 0, "stale": false, "warning": null}}
```


GET /api/hotspots

Item	Value
Purpose	Observed activity clusters: 2° cells with ≥ 3 events of M4.5+ in the last 30 days
Parameters	`limit` integer 1–50 (default 12)
Success	200: `hotspots[]` with `latitude` , `longitude` (mean of member events), `cell{lat_min,lat_max,lon_min,lon_max}` , `event_count` , `max_mag` , `latest_time` , `median_depth_km` , `model_probability` (number or `null`); plus `definition` , `window_days` , `min_mag` , `cell_deg` , `catalog_events` , `model_status` , `fetched_at` , `cache`
Errors	400 for a bad `limit` ; 502/504 if the context catalogue cannot be fetched and no copy is cached
Source	`app.py` → `hotspots()`

Cells are ranked by event count, then by largest magnitude. `model_probability` is the conditional severity score at the cluster centroid and median depth. It is `null` , never a placeholder, when the model is not loaded.

GET /api/predict

Item	Value
Purpose	Conditional severity score at a point
Parameters	`lat` −90–90 (required, alias `latitude`); `lon` −180–180 (required, aliases `lng`, `longitude`); `depth` 0–700 km (default 10)
Success	200: `probability`, `base_rate`, `relative_to_base_rate`, `band`, `features{…}`, `interpretation`, `model{trained_at,test_roc_auc}`, `context{fetched_at,stale}`
Errors	400 `missing_parameter` / `invalid_parameter` / `out_of_range`; 503 `model_unavailable`; 503 `context_unavailable`
Source	`app.py` → `predict()`

Real response for `?lat=38.3&lon=142.4&depth=30` (off north-east Japan):


```
{"probability": 0.2834, "base_rate": 0.2515, "relative_to_base_rate": 1.127, "band": "near_base_rate",
 "features": {"latitude": 38.3, "longitude": 142.4, "depth_km": 30.0,
              "local_density_30d": 1, "plate_distance_km": 134.1},
 "interpretation": "If an M4.5+ earthquake were recorded at this point and depth, the model's estimated chance that it
would be M5.0+. This is not a probability that an earthquake will occur.",
 "model": {"trained_at": "2026-10-18T21:35:45Z", "test_roc_auc": 0.5703},
 "context": {"fetched_at": "2026-10-18T21:37:35Z", "stale": false}}
```


GET /api/model-info, /api/health, /api/plates, /

- `/api/model-info` returns `ModelBundle.describe()`: the status, the load error (if any), the feature list, the architecture, and the training, split, metrics and baselines sections from `model_meta.json`.
- `/api/health` returns `{"status": "ok"|"degraded", "model_loaded", "model_error", "plate_segments", "context_cache_age_seconds", "time"}`. It reports `degraded` when the model or plate data is missing. It never calls USGS, so it stays fast and cheap.
- `/api/plates` serves `data/tectonic_plates.geojson` with `Cache-Control: max-age=86400`.
- `/` and `/dashboard-v2` render `templates/index.html`.

3.4 Request parsing and validation

Every numeric query parameter goes through one function, so validation rules are consistent across endpoints:

app.py

```python
def parse_number(name, default=None, lo=None, hi=None, cast=float, required=False, aliases=()):
    raw = request.args.get(name)
    for alias in aliases:
        raw = raw if raw not in (None, "") else request.args.get(alias)
    if raw in (None, ""):
        if required:
            raise ApiError(f"Missing required query parameter '{name}'", code="missing_parameter")
        return default
    try:
        value = cast(raw)
    except (TypeError, ValueError):
        raise ApiError(f"Parameter '{name}' must be a {'whole number' if cast is int else 'number'}",
                       code="invalid_parameter") from None
    if isinstance(value, float) and not math.isfinite(value):
        raise ApiError(f"Parameter '{name}' must be a finite number", code="invalid_parameter")
    if (lo is not None and value < lo) or (hi is not None and value > hi):
        raise ApiError(f"Parameter '{name}' must be between {lo} and {hi}", code="out_of_range")
    return value
```

Line by line:

- `request.args.get(name)` reads the query string. Aliases let `lon`, `lng` and `longitude` all work.
- An empty or missing value either fails (`required=True`) or returns the default.

- `cast(raw)` converts the text. `int("1.5")` raises, so `days=1.5` is rejected rather than silently truncated to 1.
- `float("nan")` and `float("inf")` succeed in Python, so they need the explicit `math.isfinite` check. Without it, `lat=nan` would slip past the range comparison, because every comparison with NaN is `False`.
- The range check runs last and reports the allowed bounds.
- `raise ... from None` hides the internal `ValueError` traceback, because the user-facing message is already complete.

3.5 Error handling and logging

app.py

```
@app.errorhandler(ApiError)
def _api_error(exc):
    return error_response(exc.message, exc.status, exc.code)


@app.errorhandler(UpstreamError)
def _upstream_error(exc):
    return error_response(str(exc), exc.status, exc.code)
...
@app.errorhandler(Exception)
def _unhandled(exc):
    log.exception("Unhandled error on %s", request.path)
    return error_response("Internal server error", 500, "internal_error")
```

Three kinds of failure map to three handlers:

- **`ApiError`** covers problems with the request (400) or the server's own readiness (503).
- **`UpstreamError`** covers USGS problems. Timeouts map to 504; everything else maps to 502.
- **Anything else** is a bug. The traceback goes to the server log (`log.exception`), and the client receives a generic message, because internal details may reveal file paths or code structure. A test checks that the text of an unexpected exception never reaches the response.

A fourth handler converts Werkzeug's `HTTPException` (404, 405) into the same JSON shape for `/api/*` paths. HTML paths keep the default pages.

3.6 Caching

`TTLCache` is an `OrderedDict` behind a `threading.Lock`. It is bounded to 32 entries (least recently used are evicted first) and stores `(stored_at, value)`. `cached_fetch` adds the policy:

app.py

```
    stale_max = config.STALE_MAX_AGE if stale_max is None else stale_max
    cached, age = feed_cache.get(key)
    if cached is not None and age < ttl:
        return cached, {"hit": True, "age_seconds": int(age), "stale": False, "warning": None}
    with _lock_for(key):
        cached, age = feed_cache.get(key)          # another request may have refreshed it meanwhile
        if cached is not None and age < ttl:
            return cached, {"hit": True, "age_seconds": int(age), "stale": False, "warning": None}
        try:
            payload = fetch()
        except UpstreamError as exc:
            if cached is not None and age < stale_max:
                log.warning("Serving stale cache for %s after upstream failure: %s", key, exc)
                return cached, {"hit": True, "age_seconds": int(age), "stale": True,
                                "warning": f"USGS is currently unavailable ({exc}). Showing data from "
                                           f"{int(age // 60)} min ago."}
            raise
        feed_cache.put(key, payload)
    return payload, {"hit": False, "age_seconds": 0, "stale": False, "warning": None}
```

The design has four properties:

- **Fresh hits** (feed younger than 60 s, context younger than 10 min, the 400-day analytical history younger than 1 h) never touch USGS. This protects the upstream service from a dashboard left open in many tabs.
- **Stale fallback.** If USGS fails, a copy up to 30 minutes old (6 hours for the analytical history, `stale_max`) is served, but it is *labelled*: the response has `cache.stale = true` and a human-readable warning, which the UI shows. Old data is never presented as fresh.
- **No fabrication.** With no usable copy, the error propagates and the user sees an error. The cache never invents an empty "success".
- **Single flight.** `_lock_for(key)` gives every cache key its own lock, and the cache is checked again after acquiring it. Ten simultaneous cold requests for the forecast therefore trigger one 400-day USGS download, not ten.

Derived results (sequences, forecasts, anomaly tests, aftershock fits) go into a second bounded cache, `analysis_cache`, through `memoized(key, compute)`. Their keys contain every parameter *and* the `fetched_at` of the data they were computed from, so a result can never outlive or mix data versions.

3.7 Data-source client

`usgs_client._get()` wraps `requests.get` with a timeout and a retry policy:

usgs_client.py

```
    for attempt in range(retries + 1):
        try:
            resp = http.get(url, params=params, timeout=timeout)
        except requests.exceptions.Timeout:
            last_error = UpstreamError("USGS request timed out", status=504, code="upstream_timeout")
        except requests.exceptions.RequestException as exc:
            last_error = UpstreamError(f"Could not reach USGS ({type(exc).__name__})", code="upstream_unreachable")
        else:
            if resp.status_code == 200:
                try:
                    return resp.json()
                except ValueError:
                    raise UpstreamError("USGS returned a response that is not valid JSON",
                                        code="upstream_malformed") from None
...
            if resp.status_code == 400:
                # Our query was rejected; retrying will not help.
                raise UpstreamError("USGS rejected the query parameters", status=502, code="upstream_bad_request")
            if resp.status_code == 429 or resp.status_code >= 500:
```

Which failures are retried:

- **Retried** (`TERRAWATCHER_USGS_RETRIES`, default 1): timeouts, connection errors, HTTP 429 and HTTP 5xx. The delay honours a numeric `Retry-After` header, capped at 5 s, and otherwise backs off linearly (1 s, 2 s, …).
- **Not retried:** HTTP 400 (our query is wrong, so repeating it cannot help) and invalid JSON.
- The error message includes only the exception *type* (`ConnectionError`), not its full text, which can contain URLs or internal network details.

`fetch_events()` builds the query with **full UTC timestamps**. This fixed a real defect: the original code sent dates only, which silently cut off the most recent day of data (see §14.6).

3.8 Model loading and inference in the backend

`ModelBundle` loads all three artifacts or none. `predict()` and `hotspots()` call `score_points()`:

app.py

```
def score_points(lats, lons, depths, ref_ms, context_events):
    """Build model features for arbitrary points at reference time ref_ms and score them."""
    cat_t = [e["time"] for e in context_events]
    cat_lat = [e["latitude"] for e in context_events]
    cat_lon = [e["longitude"] for e in context_events]
    n = len(lats)
    dens = F.causal_density(cat_t, cat_lat, cat_lon, [ref_ms] * n, lats, lons)
    pdist = F.plate_distance_km(lats, lons, plate_segments)
    X = F.build_feature_matrix(lats, lons, depths, dens, pdist)
    return bundle.predict_proba(X), dens, pdist
```

The reference time `ref_ms` is the moment the context catalogue was fetched, not “now”. That keeps the density window aligned with the data actually held. If a 9-minute-old cached catalogue is used, the window is the 30 days before that fetch.

3.9 Security headers and remaining limits

`_security_headers()` adds `X-Content-Type-Options: nosniff` and `Referrer-Policy: same-origin` to every response. HTML pages also get a Content-Security-Policy that allows scripts only from the app itself and `unpkg.com`, images from Esri tiles, fonts from Google Fonts, and API calls only to the same origin. The CSP allows inline *styles* (`'unsafe-inline'` in `style-src`), because the dashboard sets bar widths and colours through `style` attributes. It does not allow inline *scripts*.

Known limits (not fixed, documented in Chapter 12): the Werkzeug development server is not a production server, there is no rate limiting of incoming requests, and the caches are per process.

4 Frontend and user interface

4.1 Structure

The page is one Jinja template, `templates/index.html`, with no server-side logic beyond `url_for('static', ...)`. Scripts load in this order:

1. **Leaflet 1.9.4 from unpkg**, with Subresource Integrity (`integrity="sha256-…"`). If the CDN served modified code, the browser would refuse to run it.
2. `static/core.js`, which defines `window.TerraCore`: pure functions with no DOM access.
3. `static/charts.js` and `static/analytics.js` (`window.TerraCharts`, `window.TerraML`): pure chart and view helpers.
4. `static/dashboard.js`, an IIFE that holds the Live Monitor state and behaviour and exposes `window.TerraApp`.
5. `static/ml.js`, the five analytical views (§8.8).

Static URLs carry `?v=<file modification time>` (the `asset()` template helper), so a browser never runs a cached script from before an update.

The page regions are a top bar (status pill, freshness, refresh), a row of module tabs, a controls sidebar (shared feed controls plus each module's form, layers, legend, source), the map, an inspector and a panel area below the map. The inspector and panel area show different content in each view. A modal dialog hosts the legacy point analysis.

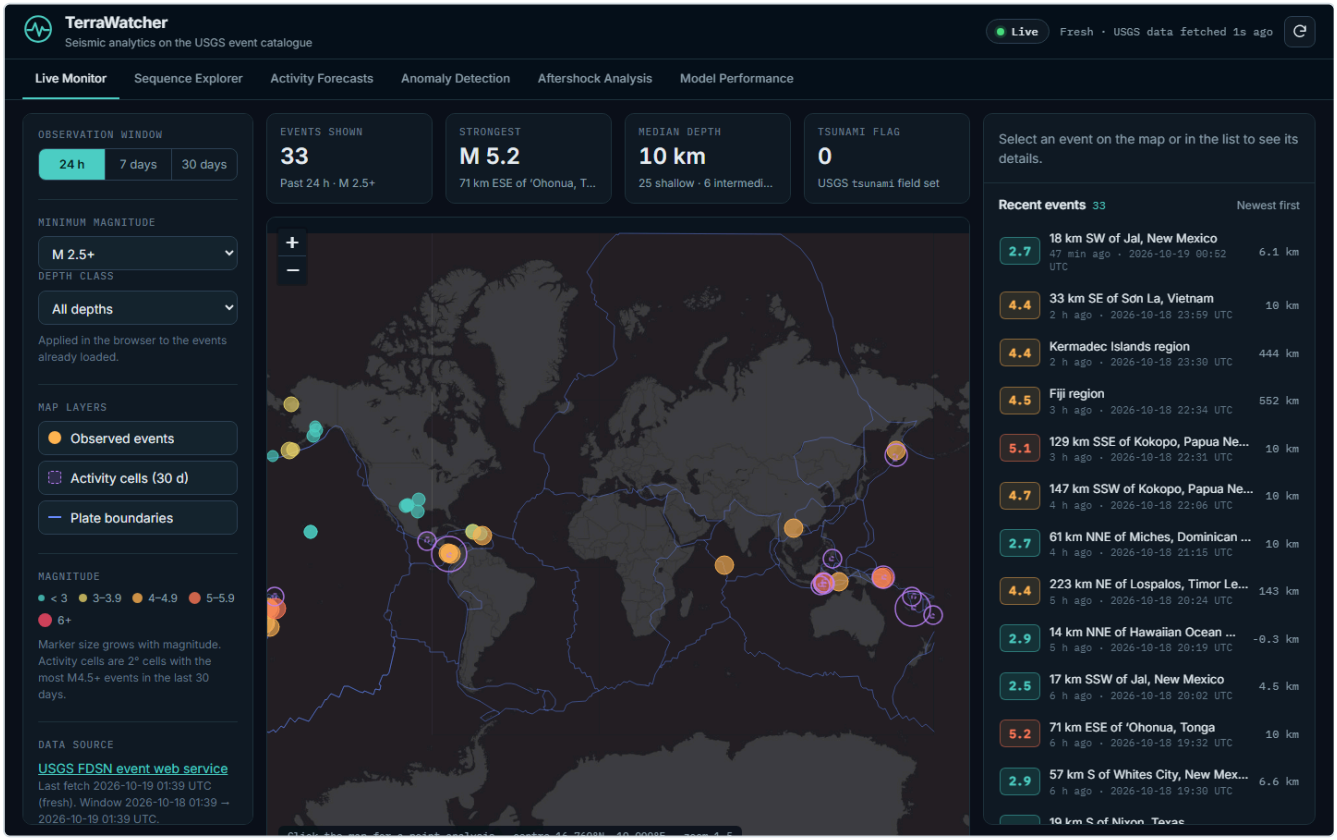


Figure 3. The Live Monitor at 1440×900 on 2026-10-19, captured with headless Chrome. The tab row above the controls leads to the five analytical views (Chapter 8). All values come from live USGS data.

4.2 State

All mutable UI state is one object, so it is easy to inspect in the browser console:

`static/dashboard.js`

```
  const state = {
    days: 1,
    minMag: 2.5,
    depth: 'all',
    allEvents: [],        // events returned by /api/earthquakes (validated)
    events: [],           // allEvents after the client-side depth filter
    feed: null,           // metadata of the last successful feed response
    hotspots: [],
    selectedId: null,
```

`allEvents` and `events` are deliberately separate. The depth filter runs in the browser on data already loaded, so changing it needs no network request. Changing `days` or `minMag` changes the query and triggers `loadFeed()`.

4.3 Networking: fetchJson

static/dashboard.js

```
  async function fetchJson(url, { signal } = {}) {
    const timeout = new AbortController();
    const timer = setTimeout(() => timeout.abort(), REQUEST_TIMEOUT_MS);
    const onAbort = () => timeout.abort();
    if (signal) signal.addEventListener('abort', onAbort);
    let response;
    try {
      response = await fetch(url, { headers: { Accept: 'application/json' }, signal: timeout.signal });
    } catch (err) {
      if (signal && signal.aborted) throw Object.assign(new Error('superseded'), { superseded: true });
      throw new Error(err.name === 'AbortError' ? 'The request timed out.' : C.errorMessage(0));
    } finally {
      clearTimeout(timer);
      if (signal) signal.removeEventListener('abort', onAbort);
    }
    let body = null;
    try { body = await response.json(); } catch (_) { /* handled below */ }
    if (!response.ok) throw Object.assign(new Error(C.errorMessage(response.status, body)), { status: response.status,
body });
    if (body === null) throw new Error('The server returned a response that is not valid JSON.');
    return body;
  }
```

What each part does:

- **Two abort sources.** A 25-second timeout, and an optional caller `signal`, used when the user changes a filter before the previous request finishes. Both feed one internal `AbortController`.
- **Superseded vs timed out.** If the caller aborted, the error is tagged `superseded`, and `loadFeed` ignores it silently, because it is not a failure the user needs to see. A real timeout produces “The request timed out.”
- **`finally`.** Always clears the timer and removes the listener, so nothing leaks across many refreshes.
- **Error bodies.** Non-2xx responses still have a JSON body (`{error, code, status}`). `C.errorMessage` extracts the server’s message, so the user sees “Parameter ‘days’ must be between 1 and 30” rather than “HTTP 400”.

4.4 Loading the feed and handling every outcome

static/dashboard.js

```
async function loadFeed() {
  if (state.feedController) state.feedController.abort();   // cancel superseded request
  const controller = new AbortController();
  state.feedController = controller;
  setStatus('loading', 'Loading…');
  $('map-loading').hidden = false;
  const params = new URLSearchParams({ days: state.days, min_mag: state.minMag });
  try {
    const data = await fetchJson(`/api/earthquakes?${params}`, { signal: controller.signal });
    const { events, rejected, malformed } = C.normalizeEvents(data);
    if (malformed) throw new Error('The event feed response was malformed.');
```

The function handles four outcomes:

Outcome	What the user sees
Success, fresh or cached	Status pill “Live”; freshness “Fresh · USGS data fetched 5s ago” or “Cached · …”
Success, stale (`cache.stale`)	Amber pill “Showing stale data”; banner with the server’s warning
Failure, earlier data exists	Red pill “Feed unavailable”; banner “… Previously loaded events are still shown.”; Retry button
Failure, no earlier data	As above, plus “Could not load earthquakes” in the list; metrics show zeros

The `finally` block hides the loading indicator *only if this request is still the current one*. A superseded request finishing late cannot hide the spinner of the newer request.

4.5 Validating responses in the browser

`static/core.js`

```
function normalizeEvents(payload) {
  const raw = payload && Array.isArray(payload.events) ? payload.events : null;
  if (!raw) return { events: [], rejected: 0, malformed: true };
  const events = [];
  let rejected = 0;
  raw.forEach((e, i) => {
    const lat = toNum(e && e.latitude);
    const lon = toNum(e && e.longitude);
    const time = toNum(e && e.time);
    if (!isValidCoord(lat, lon) || time === null) { rejected += 1; return; }
```

The server already validates, so why validate again? Because the browser should not trust that every response is well formed: a proxy error page, a partial response or a future API change could break the assumptions. `normalizeEvents` also:

- converts `depth_km` to a number or `null` , so a missing depth shows as “Not reported” rather than “NaN km”;
- keeps `url` only if it starts with `https://earthquake.usgs.gov/` , so a malformed record cannot inject a `javascript:` link into the “USGS event page” button;
- returns `malformed: true` when `events` is missing or not an array, which `loadFeed` turns into an error.

All text inserted with `innerHTML` passes through `escapeHtml` , so place names containing `<` or quotes cannot break the markup.

4.6 Map initialisation, layers and markers

`static/dashboard.js`

```
  if (state.map) return;  // guard against double initialisation
  const map = L.map('map', {
    preferCanvas: true,
    worldCopyJump: false,
    minZoom: 1,
    maxZoom: 12,
    zoomSnap: 0.5,
    maxBounds: [[-90, -200], [90, 200]],
    maxBoundsViscosity: 0.8,
    zoomControl: true,
  }).fitBounds([[-58, -170], [72, 190]]);
```

- `preferCanvas: true` draws circle markers on one `<canvas>` instead of one SVG element each, which matters with up to 2000 events.
- `fitBounds([[-58, -170], [72, 190]])` frames the inhabited latitudes and keeps the Pacific in one piece (east of 180° is allowed up to 200°). Leaflet expects `[lat, lon]` pairs.
- The guard `if (state.map) return;` prevents double initialisation, which in Leaflet throws “Map container is already initialized”.

There are three `L.layerGroup`s, created in drawing order: plates (bottom), clusters, events (top). The layer toggle buttons add a group to the map or remove it; the data stays in memory.

static/dashboard.js

```
  // Draw small events first so large ones stay on top.
  const ordered = [...state.events].sort((a, b) => (a.mag ?? -9) - (b.mag ?? -9));
  ordered.forEach((e) => {
    const color = C.magColor(e.mag);
    const marker = L.circleMarker([e.latitude, e.longitude], {
      radius: C.magRadius(e.mag), color, weight: 1, fillColor: color, fillOpacity: 0.65, bubblingMouseEvents: false,
    });
```

`bubblingMouseEvents: false` stops a marker click from also triggering the map’s click handler, which would open point analysis at the same time. Markers are kept in `state.markers` (a `Map` from event id to marker) so that selecting an event in the list can restyle the matching marker.

Clustering. TerraWatcher does *not* use marker clustering (such as Leaflet.markercluster). Markers are drawn individually. “Activity clusters” are a server-side grid aggregation (§5.5), not a rendering technique.

4.7 Filtering, selection and the detail view

- **Window and magnitude** change `state.days` / `state.minMag` and call `loadFeed()` (server round trip).
- **Depth class** changes `state.depth` and calls `applyDepthFilter()`, which recomputes `state.events` and re-renders the metrics, list, detail and markers without any network request. If the selected event is filtered out, the selection is cleared.
- **Selecting** an event, from a marker or a list row, calls `selectEvent(id, {pan})`. It sets `state.selectedId`, re-renders the detail panel, updates `aria-pressed` on list rows, gives the selected marker a white outline and, when selected from the list, flies the map to it.

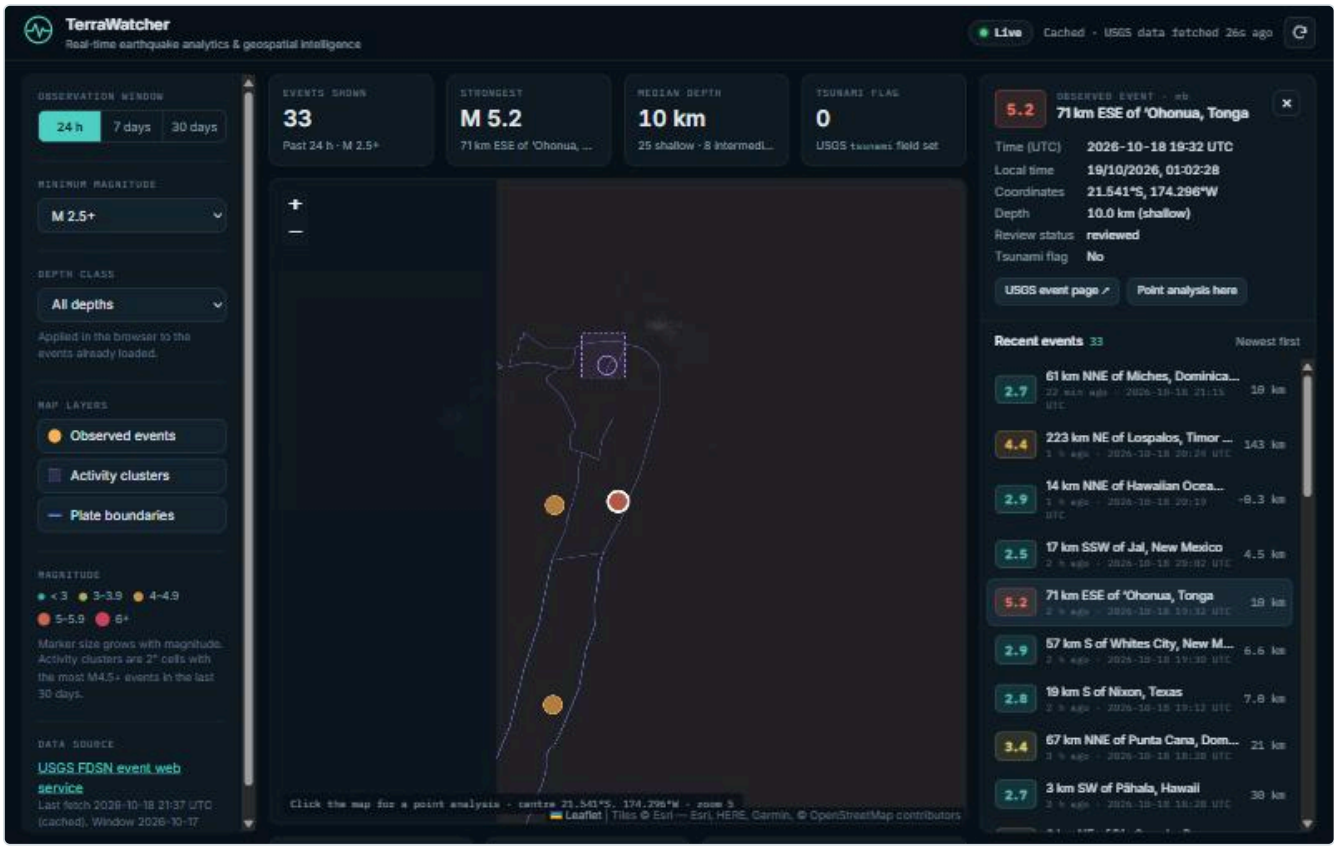


Figure 4. A selected event. During acceptance testing every field in the panel was compared with the raw `/api/earthquakes` record for the same id: place, magnitude, depth 10.0 km, coordinates 21.541°S 174.296°W and time.

4.8 Dashboard metrics

`core.summarize(events)` computes every number shown in the metric cards and charts from the events currently displayed. The values are never hard-coded:

static/core.js

```
function summarize(events) {
  const mags = events.map((e) => e.mag).filter(Number.isFinite);
  const depths = events.map((e) => e.depth).filter(Number.isFinite);
  let strongest = null;
  events.forEach((e) => { if (Number.isFinite(e.mag) && (!strongest || e.mag > strongest.mag)) strongest = e; });
```

Events with a missing magnitude still count in “Events shown”, but they are excluded from the magnitude histogram, and the `withMagnitude` count tells the chart when there is nothing to plot. The median is used for depth because depth distributions are skewed: a few 600 km events would distort a mean.

4.9 Point analysis

Clicking the map (or “Point analysis here” in the detail panel) opens a modal dialog with editable latitude, longitude and depth inputs, so the feature is also usable from the keyboard. The modal shows the probability, the ratio to the base rate, every feature value, the model’s test ROC-AUC and the sentence “not a probability that an earthquake will occur”. Escape closes it and returns focus to the element that opened it.

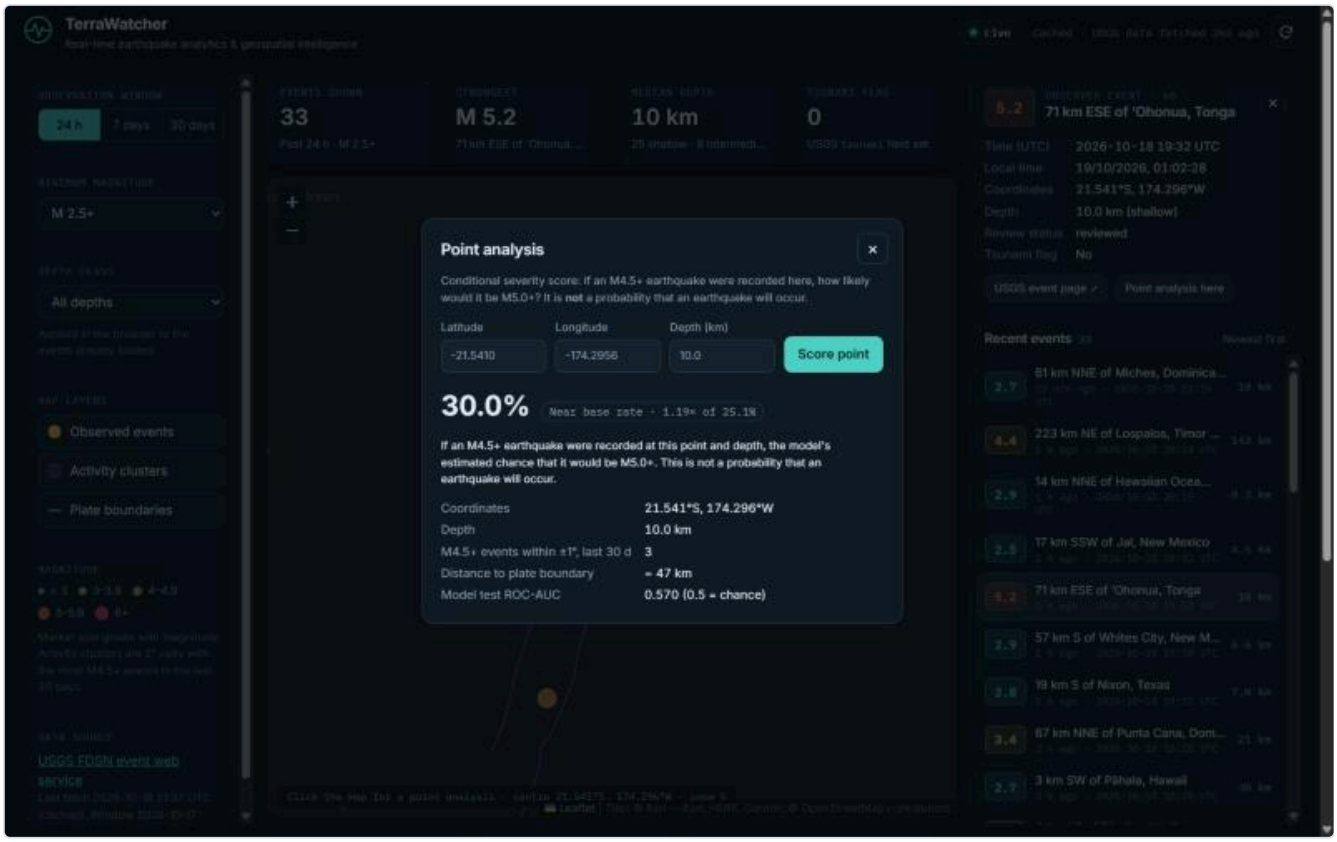


Figure 5. Point analysis for the selected Tonga event. The 30.0 % shown equals the `probability` returned by `/api/predict` for the same inputs (checked during acceptance testing).

4.10 Responsive design and accessibility

- **Above 1280 px:** three columns (controls · map · inspector). From 860 to 1280 px: two columns, with the inspector placed under the map. Below 860 px: one column, and the controls become a slide-in drawer opened by the ☰ button, closable with Escape.
- `display: contents` on `<main>` at narrower widths lets the event list sit directly under the map, instead of after the charts, while keeping a logical DOM order.
- All controls are native `<button>` / `<select>` / `<input>` elements with labels. The time window uses `role="radiogroup"` and `aria-checked`; the layer toggles use `aria-pressed`. A “Skip to event list” link appears on keyboard focus. Status changes are announced through `role="status"` and `aria-live`.
- `prefers-reduced-motion` disables animations.

During testing at 375×812 the page had no horizontal scroll (`scrollWidth` equalled `innerWidth`).

4.11 Handling malformed or incomplete responses

Situation	Handling
Response is not JSON	`fetchJson` throws "The server returned a response that is not valid JSON."
`events` missing or not an array	`normalizeEvents` returns `malformed: true`; `loadFeed` shows an error
An event without valid coordinates or time	Dropped and counted (`console.warn`); the rest render
Missing magnitude	Badge shows "—"; grey marker of minimum size; excluded from the histogram
Missing depth	"Not reported" in the detail view; excluded from the depth statistics
Missing place	"Location unavailable"
`/api/hotspots` or `/api/plates` fails	A toast explains it; the event layer keeps working
Leaflet fails to load	The map area explains it; list, metrics and charts still work

4.12 End-to-end trace: from a click to pixels

1. The user clicks **7 days**. The handler sets `aria-checked`, sets `state.days = 7` and calls `loadFeed()`.
2. `loadFeed` aborts any in-flight feed request and calls `fetchJson('/api/earthquakes?days=7&min_mag=2.5', {signal})`.
3. Flask's `earthquakes()` validates the parameters, misses the cache and calls `usgs_client.fetch_events(now − 7 d, now, 2.5, 2000)`.
4. USGS returns GeoJSON. `normalize_collection` validates and de-duplicates it. The payload is cached and returned with `cache.hit = false`.
5. The browser receives the JSON (during testing: 337 events). `normalizeEvents` validates it, `applyDepthFilter` keeps all 337 (depth "all"), and `renderMetrics` shows **337** and "Past 7 days · M 2.5+". The strongest event card showed **M 6.9**, which matched `Math.max` over the API magnitudes.
6. `renderMarkers` clears the event layer and draws 337 canvas circles; `renderList` shows the newest 200.

5 Geospatial concepts

5.1 Latitude and longitude

A point on Earth is located by two angles:

- **Latitude** (φ) measures north–south position, from −90° (South Pole) through 0° (Equator) to +90° (North Pole).
- **Longitude** (λ) measures east–west position from the Greenwich meridian, from −180° (west) to +180° (east). The line at ±180° is the **antimeridian**. It runs through the Pacific, past Fiji, Tonga and the Aleutian Islands, which are some of the most seismically active places on Earth.

USGS reports coordinates in the WGS84 datum, the same one GPS uses. One degree of latitude is always about 111.2 km (TerraWatcher uses `KM_PER_DEG = 111.195`). One degree of longitude shrinks with latitude: it is 111.2 km × cos φ, so about 78.6 km at 45° and 0 km at the poles.

5.2 Projections and the map

The map uses the **Web Mercator** projection (Leaflet's default, EPSG:3857). It preserves shapes locally but enlarges areas towards the poles, so Greenland looks as large as Africa. This matters for interpretation: a cluster of markers at 60°N covers less ground than an identical-looking cluster at the Equator. TerraWatcher's numerical features do *not* use Mercator. They work directly with degrees and an approximate local scale correction (§5.6).

5.3 From a data record to a marker

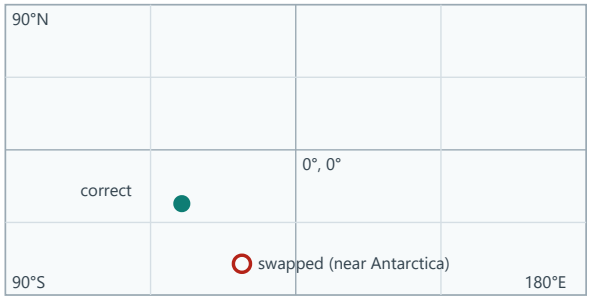
Each USGS event has an **epicentre**, the point on the surface above where the rupture began, plus a **depth** below the surface. The epicentre plus depth is the **hypocentre**. TerraWatcher plots the epicentre and shows depth as text and as a filter, because a 2-D map cannot show depth directly.

The pipeline from record to pixel:

1. USGS: `"geometry": {"coordinates": [−174.2956, −21.541, 10.0]}`, i.e. **[longitude, latitude, depth]**.
2. `normalize_feature`: `lon, lat = coords[0], coords[1]`, stored as named fields `latitude: −21.541`, `longitude: −174.2956`, `depth_km: 10.0`.
3. Browser: `L.circleMarker([e.latitude, e.longitude])`. Leaflet expects **[latitude, longitude]**.
4. Leaflet projects the point to Web Mercator pixels for the current zoom and draws it on the canvas.

5.4 The coordinate-order trap

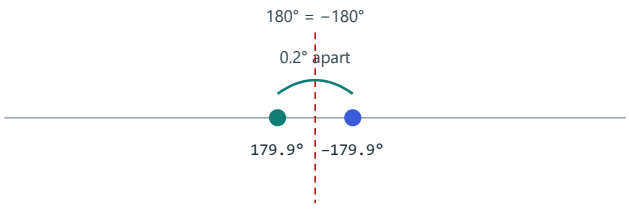
Coordinate order: GeoJSON vs Leaflet




```
USGS: coordinates [-70.6, -33.4, 35.0] // lon, lat, depth
Leaflet: L.circleMarker([lat, lon])
```

Converted once, by name, in normalize_feature();
the browser only sees named latitude/longitude fields.

Longitude wrap at the antimeridian (±180°)




```
naive: |179.9 − (−179.9)| = 359.8° ✗
wrapped: ((Δ + 180) mod 360) − 180 = −0.2° ✓
```

features.wrap_lon() is applied in both plate_distance_km()
and causal_density(), so Fiji/Tonga/Aleutian events near ±180°
get correct distances and neighbour counts.

Figure 6. Left: GeoJSON stores `[lon, lat]`, Leaflet takes `[lat, lon]`. Swapping them moves a Santiago, Chile earthquake to the ocean near Antarctica. Right: why longitude differences must be wrapped.

Swapping latitude and longitude is the most common geospatial bug. Its consequences depend on the values:

- If |longitude| > 90, the swapped "latitude" is invalid. Leaflet may silently clamp it or draw the marker off the map.
- If both values are within ±90 (much of Europe, Africa and South America), the swapped point is valid but **wrong**, and nothing warns you. Santiago (−33.4, −70.6) moves to (−70.6, −33.4), off Antarctica.

TerraWatcher avoids the trap structurally. The array is unpacked exactly once, in `normalize_feature`, and everything downstream uses named fields (`latitude`, `longitude`). A unit test (`test_normalize_valid_feature_maps_fields`) asserts that `[142.5, 38.1]` becomes `latitude 38.1`, `longitude 142.5`.

5.5 Grids and spatial aggregation

Activity clusters bin events into **2° × 2° cells**. The cell index is `floor(lat / 2)` and `floor(lon / 2)`, so the cell containing (38.3°N, 142.2°E) is latitude 38–40 and longitude 142–144. Cells are equal in *degrees*, not in area: a 2° cell at 60° latitude covers half the ground of one at the Equator (cos 60° = 0.5). For a visual summary of where activity is concentrated this distortion is acceptable. For any density *comparison* between regions it would need correcting.

On the map each cluster is drawn as a dashed rectangle (the true cell) plus a ring at the mean position of its events. The ring radius grows with √(event count), because area should scale with count. The `/api/hotspots` definition text and the popup both say "observed activity, not a forecast".

> **IMPORTANT** A cell with many recent earthquakes is an observation about the past. Some of that activity reflects aftershock sequences, which do decay over time and are studied statistically. But a visual concentration of historical events does not, by itself, establish a reliable prediction of where future earthquakes will occur. Large earthquakes also happen in areas that were quiet in the preceding month.

5.6 Distances: degrees, kilometres and the antimeridian

`plate_distance_km` needs the distance from a point to a line segment. The exact great-circle computation is costly. Instead, TerraWatcher projects each segment into a **local equirectangular frame** centred on the query point:

- $x = \Delta\lambda \times \cos \varphi_{point}$, with $\Delta\lambda$ wrapped into [−180°, 180°)
- $y = \Delta\varphi$

Then it applies ordinary 2-D point-to-segment geometry. The point projects onto the segment at parameter t, clamped to [0, 1], and the distance to that closest point is multiplied by 111.195 km per degree:

features.py

```
        k = np.cos(np.radians(plat))
        ax, ay = wrap_lon(lon1 - plon) * k, lat1 - plat   # segment start relative to point
        bx, by = wrap_lon(lon2 - plon) * k, lat2 - plat   # segment end relative to point
        dx, dy = bx - ax, by - ay
        l2 = dx * dx + dy * dy
        with np.errstate(invalid="ignore", divide="ignore"):
            t = np.where(l2 > 0, -(ax * dx + ay * dy) / l2, 0.0)
        t = np.clip(t, 0.0, 1.0)
        cx, cy = ax + t * dx, ay + t * dy                  # closest point on segment
        out[start:start + chunk] = np.sqrt(cx * cx + cy * cy).min(axis=1) * KM_PER_DEG
```

Each line, explained:

- The point sits at the origin, so segment endpoints are expressed *relative to it* (`ax, ay`, `bx, by`).
- `wrap_lon` turns a 359.8° difference into −0.2°, so a point at 179.9° and a segment at −179.9° are 0.2° apart, not 359.8° (tested in `test_plate_distance_across_antimeridian`).
- `t = −(a·d)/|d|²` is the projection of the origin onto the line through the segment. Clipping keeps it on the segment. A zero-length segment (`l2 = 0`) uses its start point.
- The computation is vectorised over 512 points × 6051 segments at a time (about 3 million distances per chunk), then `.min(axis=1)` picks the nearest segment.

Accuracy: good for distances of tens to a few hundred kilometres away from the poles, which is the range that matters for "near a plate boundary". It degrades over long distances because the scale factor cos φ is taken at the query point. No formal error bound was computed.

5.7 Zoom levels, tiles and layers

At zoom z the world is 256 × 2^z pixels wide. TerraWatcher allows zoom 1–12 with half-step snapping. Basemap **tiles** are 256-pixel images fetched on demand from Esri's World Dark Gray Canvas service (URL pattern `…/tile/{z}/{y}/{x}`). The app draws three overlay **layers** above the tiles: plate boundaries, activity clusters and observed events. Each is an `L.layerGroup` that the sidebar toggles add to or remove from the map. No marker clustering library is used.

6 Earthquake data pipeline

6.1 Source

The only earthquake data source is the **USGS FDSN event web service**, `https://earthquake.usgs.gov/fdsnws/event/1/`. It is public, needs no API key and follows the International Federation of Digital Seismograph Networks (FDSN) standard. TerraWatcher uses two methods:

- `query`, with `format=geojson`, `starttime`, `endtime`, `minmagnitude`, `orderby=time`, `limit` and `eventtype=earthquake` (which excludes quarry blasts and explosions);
- `count`, called only when a query returns exactly `limit` events, to report how many exist in total.

The catalogue merges contributions from many regional networks (the `net` field: `us`, `ci`, `ak`, …). That is why magnitude types vary (`mb`, `ml`, `Mww`, `md`) and why small events are much better recorded in some regions than others.

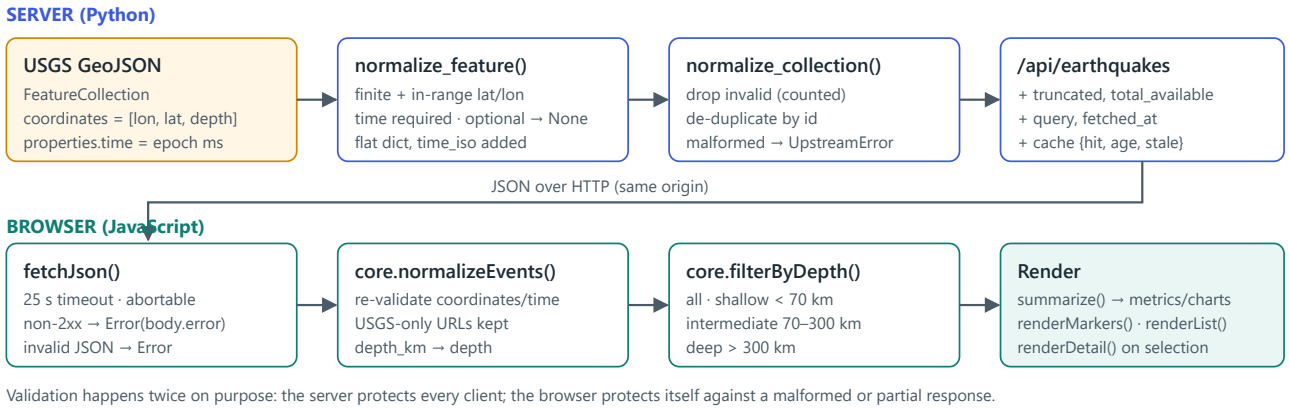


Figure 7. Every transformation an event goes through between USGS and the screen.

6.2 HTTP, timeouts and failure handling

Condition	Behaviour	Status returned to browser
Timeout (15 s default)	Retry once, then fail	504 `upstream_timeout`
Connection error / DNS failure	Retry once, then fail	502 `upstream_unreachable`
HTTP 429	Wait `Retry-After` (≤ 5 s), retry	502 `upstream_rate_limited`
HTTP 5xx	Back off 1 s, retry	502 `upstream_server_error`
HTTP 400	No retry	502 `upstream_bad_request`
200 but not JSON, or no `features` list	No retry	502 `upstream_malformed`
Any failure while a copy ≤ 30 min old is cached	Serve that copy	200 with `cache.stale = true`

6.3 Parsing and normalisation

`usgs_client.py`

```
    lon, lat = _finite(coords[0]), _finite(coords[1])
    if lon is None or lat is None or not (-90 <= lat <= 90) or not (-180 <= lon <= 180):
        return None
    depth = _finite(coords[2]) if len(coords) > 2 else None

    t = _finite(props.get("time"))
    if t is None:
        return None
    time_ms = int(t)
```

`_finite()` converts a value to a float and returns `None` for `None`, non-numbers, NaN and infinity. Three outcomes follow:

- **Required fields** (coordinates and time): if missing or invalid, the whole record is dropped (`return None`) and counted in `dropped_invalid`.
- **Optional fields** (depth, magnitude, place and the rest) pass through as `None`. They are never replaced with defaults. The original code substituted 0.0 for missing coordinates and 10 km for missing depth, inventing data.
- A **2-element coordinate array** is valid (depth unknown); a 1-element array is not.

6.4 Identifiers and duplicates

usgs_client.py

```
    events, seen = [], set()
    dropped = duplicates = 0
    for feature in payload["features"]:
        event = normalize_feature(feature)
        if event is None:
            dropped += 1
            continue
        if event["id"] in seen:
            duplicates += 1
            continue
        seen.add(event["id"])
        events.append(event)
```

The event id is the GeoJSON `id` (for example `us6000u37h`). If it is absent, the first entry of `properties.ids` is used, and failing that a synthetic `"lat,lon,time"` string. The first occurrence of each id is kept. Training also de-duplicates across download chunks by keying a dictionary on id.

6.5 Field provenance: source, calculated or model-generated

Field	Origin	Notes
`latitude`, `longitude`, `depth_km`	**Source** (`geometry.coordinates`)	renamed, validated
`mag`, `mag_type`, `place`, `status`, `tsunami`, `alert`, `felt`, `sig`, `net`, `url`, `updated`	**Source** (`properties`)	passed through (`tsunami` → boolean)
`time`	**Source**	epoch milliseconds, UTC
`time_iso`	**Calculated**	ISO-8601 rendering of `time`
`count`, `truncated`, `total_available`, `dropped_invalid`, `duplicates_removed`	**Calculated**	per response (`total_available` from the USGS count endpoint when truncated)
`fetched_at`, `cache.*`	**Calculated**	freshness metadata
Metric cards, histograms	**Calculated** (browser)	from displayed events
Activity clusters: `event_count`, `max_mag`, `median_depth_km`, centroid	**Calculated**	from the 30-day M4.5+ catalogue
`local_density_30d`, `plate_distance_km`	**Calculated**	model features
`probability`, `model_probability`, `relative_to_base_rate`, `band`	**Model-generated**	exploratory

Worked example: one record

USGS sends (abridged):


```
{"type": "Feature", "id": "us6000u37h",
 "properties": {"mag": 5.2, "place": "71 km ESE of ‘Ohonua, Tonga", "time": 1792351948734,
                "updated": 1792354380040, "tsunami": 0, "magType": "mb", "type": "earthquake",
                "status": "reviewed", "net": "us", "sig": 416,
                "url": "https://earthquake.usgs.gov/earthquakes/eventpage/us6000u37h"},
 "geometry": {"type": "Point", "coordinates": [-174.2956, -21.541, 10.0]}}
```


TerraWatcher’s API returns `{"id": "us6000u37h", "time": 1792351948734, "time_iso": "2026-10-18T19:32:28.734Z", "latitude": -21.541, "longitude": -174.2956, "depth_km": 10.0, "mag": 5.2, "mag_type": "mb", "tsunami": false, …}`. In the browser, `normalizeEvents` turns this into `{id, time, latitude, longitude, depth: 10, mag: 5.2, magType: "mb", place, status: "reviewed", tsunami: false, url}`. That renders as an orange-red marker (M5–5.9) with radius 2.2 + 5.2 × 1.7 ≈ 11 px and as “10.0 km (shallow)” in the detail panel.

6.6 Time handling

- USGS `time` is **milliseconds since 1970-01-01 UTC**. Python divides by 1000 for `datetime`; JavaScript’s `new Date(ms)` takes milliseconds directly.
- Query windows are built from `datetime.now(timezone.utc)` and sent as `YYYY-MM-DDTHH:MM:SS` (UTC).
- The UI shows both UTC (`formatUtc`) and the viewer’s local time, and relative times (“22 min ago”).

6.7 Refresh intervals and freshness

Mechanism	Interval
Browser auto-refresh (`REFRESH_MS`)	every 3 minutes, skipped while the tab is hidden
Feed cache (`FEED_CACHE_TTL`)	60 seconds per `(days, min_mag)`
Context catalogue cache (`CONTEXT_CACHE_TTL`)	10 minutes
Stale fallback limit (`STALE_MAX_AGE`)	30 minutes
Freshness label re-render	every 30 seconds

The top bar always says which of three states applies: **Fresh** (fetched from USGS for this request), **Cached** (served from the 60 s cache) or **Stale copy** (USGS failed; old data clearly labelled). The sidebar shows the exact fetch time and the query window. TerraWatcher never substitutes fake events: with USGS unreachable and nothing cached, the list says “Could not load earthquakes” and the map shows no event markers (Figure 8).

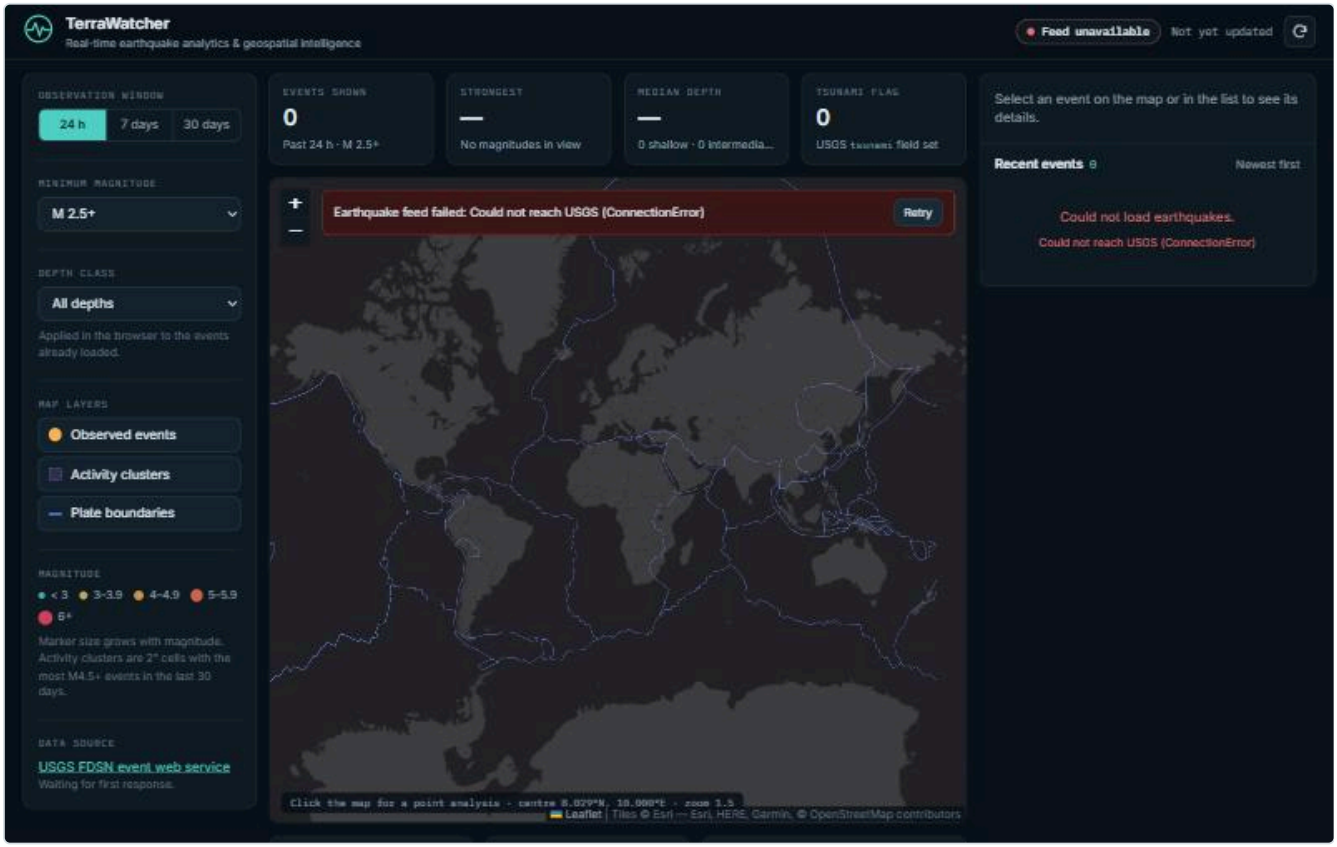


Figure 8. A real failure test: a second instance started with `TERRAWATCHER_USGS_URL` pointing to an unresolvable host. The error banner and Retry button appear, the list states the cause, the plate layer and map still work, and no events are invented.

7 Legacy model: the conditional severity classifier

7.1 The question, stated precisely

> Given that an earthquake of magnitude **≥ 4.5** has been recorded, at a given latitude, longitude and depth, with a given amount of M4.5+ activity in the surrounding ±1° during the previous 30 days, and a given distance to the nearest plate boundary, what is the probability that its magnitude is **≥ 5.0**?

This is **binary classification of recorded events**. A positive label means "this already-recorded M4.5+ event was M5.0 or larger". It says nothing about whether an event will occur. When the dashboard scores an arbitrary map point, it is asking a hypothetical: *if* an M4.5+ event happened here now, how likely is it to be M5+?

Why this formulation? It is honest and testable. Every training row has a real, known answer, and the same feature definitions can be computed at serving time without any future information. It is also a hard problem with a well-known physical answer: by the Gutenberg–Richter relation (§9.1), the share of events above a magnitude threshold is roughly constant. Strong performance should therefore *not* be expected, and the evaluation confirms this.

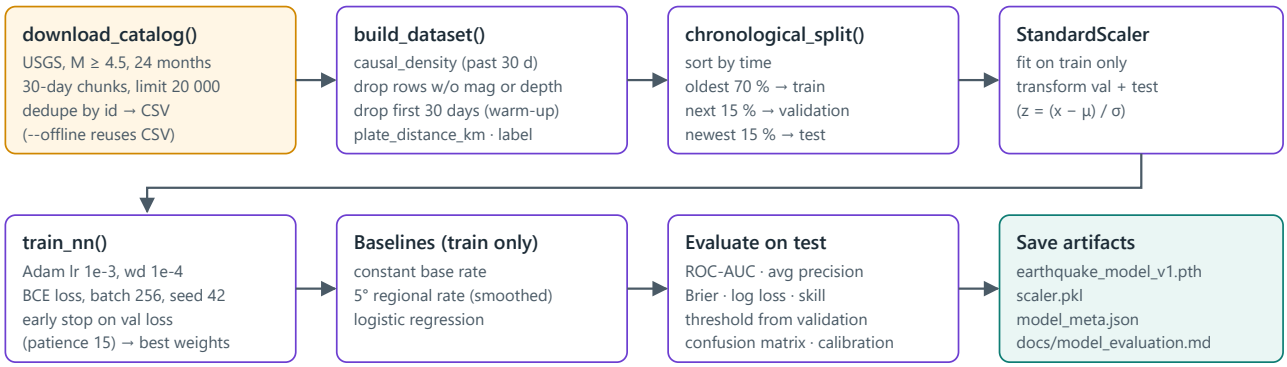


Figure 9. The offline pipeline in `train_model.py`, one box per function.

7.2 Dataset origin and schema

`download_catalog()` fetches **24 months** of M ≥ 4.5 events in 30-day chunks (`limit=20000` per chunk; the script aborts if a chunk reaches the limit, which would mean truncation). For the current artifacts: 15,855 catalogue events from 2024-10-28 to 2026-10-18. They are saved to `data/training_catalog.csv` (git-ignored) with the normalised event fields (`id, time, latitude, longitude, depth_km, mag, …`). `--offline` retrains from this file.

7.3 Features and units

#	Feature	Unit	Computation	Training mean / std*
1	`latitude`	degrees	from the event	8.85 / 33.31
2	`longitude`	degrees	from the event	62.23 / 115.28
3	`depth_km`	km	from the event	56.0 / 102.7
4	`local_density_30d`	count	`causal_density`, ±1° box, previous 30 days, M4.5+	29.5 / 73.9
5	`plate_distance_km`	km	`plate_distance_km`, nearest of 6051 segments	94.2 / 140.6

*From the fitted `scaler.pkl` (`mean_`, `scale_`). The large density spread comes from aftershock sequences: a single large earthquake can produce hundreds of M4.5+ events nearby within weeks.

7.4 Label construction and the M4.5 decision

`label = int(mag >= 5.0)`, giving a training positive rate of 25.2 %. The choice of the *catalogue* threshold, M4.5, is deliberate. Below roughly M4.5 the global catalogue is **incomplete in a region-dependent way**: California, Alaska and Japan record M2.5 events densely, while many oceanic and developing regions do not. With an M2.5 catalogue, "this event is M5+" would partly mean "this event happened where only large events get recorded", and the model would learn network coverage rather than geophysics. This is a design argument based on catalogue properties. It was **not measured** in this project.

7.5 Leakage-free feature engineering

features.py

```
    lo = np.searchsorted(ct, q_times - window_days * DAY_MS, side="left")
    hi = np.searchsorted(ct, q_times, side="left")   # strictly before q_time
    out = np.zeros(len(q_times), dtype=np.int64)
    for i in range(len(q_times)):
        if hi[i] <= lo[i]:
            continue
        dlat = np.abs(clat[lo[i]:hi[i]] - q_lats[i])
        dlon = np.abs(wrap_lon(clon[lo[i]:hi[i]] - q_lons[i]))
        out[i] = int(np.count_nonzero((dlat <= radius_deg) & (dlon <= radius_deg)))
```

- The catalogue is sorted by time once. `searchsorted` finds, for each query time t, the index range of events with t − 30 d ≤ time < t. Using `side="left"` for the upper bound excludes events *at* t, including the event itself.
- Within that slice, a box test keeps events within ±1° of latitude and ±1° of wrapped longitude.
- Cost: O(n log n) for sorting plus about the size of the 30-day slice per query, instead of comparing every pair of events in the catalogue.

`test_density_feature_does_not_see_the_future` appends 400 days of later events to a catalogue and checks that **no earlier sample's density changes**. That is the definition of no look-ahead.

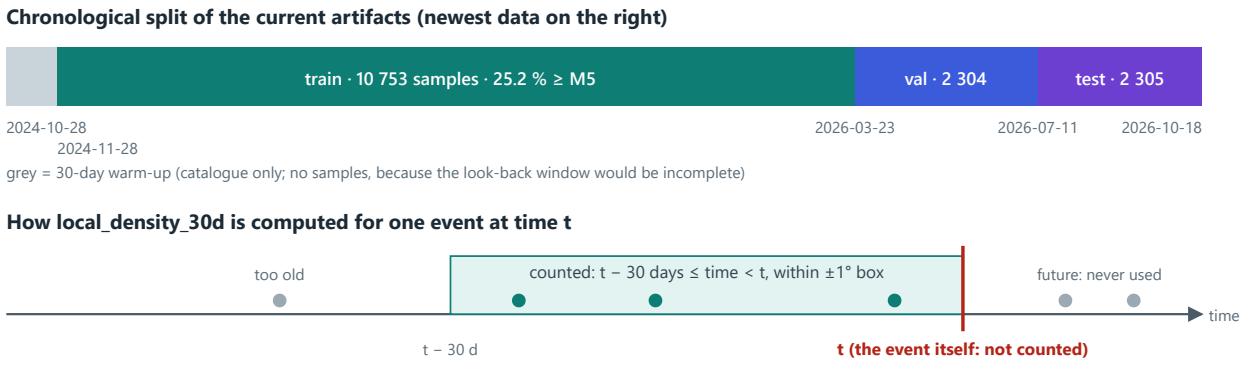


Figure 10. Top: the actual chronological split of the current artifacts. Bottom: the window counted by `local_density_30d` for one event.

7.6 Building the dataset

train_model.py

```
    # Density counts every catalogue event (even ones lacking magnitude/depth).
    cat["local_density_30d"] = F.causal_density(cat["time"], cat["latitude"], cat["longitude"],
                                                cat["time"], cat["latitude"], cat["longitude"])

    samples = cat.dropna(subset=["mag", "depth_km"]).copy()
    # Warm-up: the first 30 days have an incomplete look-back window.
    warmup_end = cat["time"].min() + F.DENSITY_WINDOW_DAYS * F.DAY_MS
    samples = samples[samples["time"] >= warmup_end].copy()
    samples["plate_distance_km"] = F.plate_distance_km(samples["latitude"], samples["longitude"], segments)
    samples["label"] = (samples["mag"] >= F.SIGNIFICANT_MAG).astype(int)
```

- Rows without magnitude (no label possible) or depth (a feature would be missing) are not samples, but they still count towards *other* events' density, because at serving time the context catalogue contains them too.
- The **warm-up** removes the first 30 days of samples. Their look-back window extends before the download start, so their density would be artificially low. 15,362 samples remain.

7.7 Train/validation/test split

train_model.py

```python
def chronological_split(df, train_frac=0.70, val_frac=0.15):
    df = df.sort_values("time").reset_index(drop=True)
    n = len(df)
    i_train, i_val = int(n * train_frac), int(n * (train_frac + val_frac))
    return df.iloc[:i_train], df.iloc[i_train:i_val], df.iloc[i_val:]
```

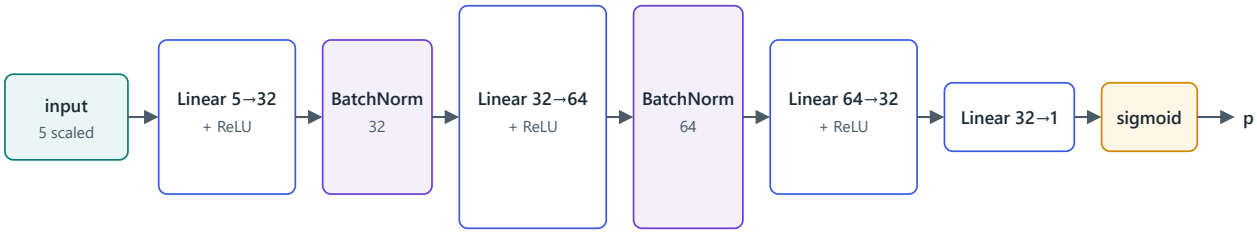
Split	Samples	Positive rate	Period
Train	10,753	25.2 %	2024-11-28 → 2026-03-23
Validation	2,304	24.1 %	2026-03-23 → 2026-07-11
Test	2,305	27.2 %	2026-07-11 → 2026-10-18

The training script asserts that the periods are in order without overlap. The validation set is used for early stopping and for choosing a decision threshold. The test set is used **once**, for reporting.

7.8 Preprocessing: standardisation

`StandardScaler` replaces each feature x with z = (x − μ) / σ, using the μ and σ of the **training split only**. Fitting on all data would leak test-period statistics into training. Standardisation matters because the raw features have very different scales (latitude ±90, density 0 to several hundred); without it, gradient descent would be dominated by the largest-valued feature. The same fitted scaler is saved to `scaler.pkl` and applied at inference.

7.9 The model: a small multilayer perceptron



4,609 trainable parameters (plus BatchNorm running statistics). Box height loosely indicates layer width.

Figure 11. `EarthquakeNN` as defined in `ml_model.py`.

Beginner's view. A neural network is a chain of simple functions. Each **linear layer** computes weighted sums of its inputs. Each **ReLU** keeps positive values and zeroes negative ones, which lets the chain represent curved, non-linear relationships. The final **sigmoid** squashes any number into (0, 1), so it can be read as a probability. Training adjusts the weights so that predicted probabilities move towards the true labels.

Formally. For a standardised input $x \in \mathbb{R}^5$:

- $h_1 = BN_{32}(ReLU(W_1x + b_1))$
- $h_2 = BN_{64}(ReLU(W_2h_1 + b_2))$
- $h_3 = ReLU(W_3h_2 + b_3)$
- $p = \sigma(w_4^\top h_3 + b_4)$, with $\sigma(z) = 1 / (1 + e^{(-z)})$

BatchNorm (BN) normalises each unit using batch statistics during training and stored running averages during inference. This is why `model.eval()` is essential: in training mode, BatchNorm cannot normalise a single input row (PyTorch raises an error for a batch of one). The network has 4,609 trainable parameters, which is small relative to 10,753 training rows.

The architecture was inherited from the original project and kept. Honestly, an MLP is not obviously the best choice: logistic regression on the same features performs almost identically (§7.12).

7.10 Training procedure

train_model.py

```
    for epoch in range(1, epochs + 1):
        model.train()
        perm = torch.randperm(len(Xt), generator=gen)
        for start in range(0, len(Xt), batch_size):
            idx = perm[start:start + batch_size]
            if len(idx) < 2:          # BatchNorm needs >1 sample in training mode
                continue
            opt.zero_grad()
            loss = loss_fn(model(Xt[idx]), yt[idx])
            loss.backward()
            opt.step()
        model.eval()
        with torch.no_grad():
            val_loss = loss_fn(model(Xv), yv).item()
        if val_loss < best_loss - 1e-5:
            best_loss, best_epoch, bad = val_loss, epoch, 0
            best_state = {k: v.clone() for k, v in model.state_dict().items()}
```

- **Loss:** binary cross-entropy, $L = -[y \log p + (1 - y) \log(1 - p)]$, averaged over the batch. It rewards confident correct probabilities and heavily penalises confident wrong ones.
- **Optimiser:** Adam (learning rate 10^{-3}, weight decay 10^{-4}), mini-batches of 256, shuffled each epoch by a seeded generator.
- **Early stopping:** after every epoch the validation loss is computed. The best weights are kept in memory; training stops after 15 epochs without improvement, and the best weights are restored. In the current run, the best epoch was **10** of 25 run (validation log loss 0.5477).
- **No class weighting.** Re-weighting positives would raise recall but distort the probabilities. Since the output is presented *as a probability*, calibration was preferred.
- **Reproducibility:** seed 42 for Python, NumPy and PyTorch. Retraining the same CSV under torch 2.5.1 and 2.14.1 produced identical test metrics. `test_training_is_reproducible_with_a_seed` checks this on synthetic data.

7.11 Serialisation and loading

Training writes four files: `earthquake_model_v1.pth` (a `state_dict`, weights only), `scaler.pkl` (joblib pickle), `model_meta.json` (schema, data, split, training settings, metrics, baselines, limitations) and `docs/model_evaluation.md` (a readable report). All four are written only after training and evaluation finish, so a failed run cannot leave a half-updated set.

ml_model.py

```
            if meta.get("feature_names") != FEATURE_NAMES:
                raise ValueError(
                    f"Feature schema mismatch: artifact has {meta.get('feature_names')}, code expects {FEATURE_NAMES}"
                )
            scaler = joblib.load(scaler_path)
            if getattr(scaler, "n_features_in_", None) != len(FEATURE_NAMES):
                raise ValueError("Scaler was fitted on a different number of features")
            model = EarthquakeNN()
            model.load_state_dict(torch.load(model_path, map_location="cpu", weights_only=True))
            model.eval()   # BatchNorm must use running statistics for inference
```

Three checks guard the contract between training and serving:

1. **The feature names and order** in `model_meta.json` must equal `FEATURE_NAMES` in code. Renaming or reordering a feature without retraining is caught at start-up. Tested with the old names (`depth` , `local_density` , `plate_distance`).
2. **The scaler width** must be 5.
3. `load_state_dict` is strict, so any architecture mismatch raises. `weights_only=True` stops a tampered `.pth` file from executing code during loading.

`scaler.pkl` is still a pickle, so it must come from a trusted source. `requirements.txt` pins scikit-learn to `<1.8` , the version family that wrote it, and `pytest.ini` turns sklearn's `InconsistentVersionWarning` into a test failure.

7.12 Evaluation results

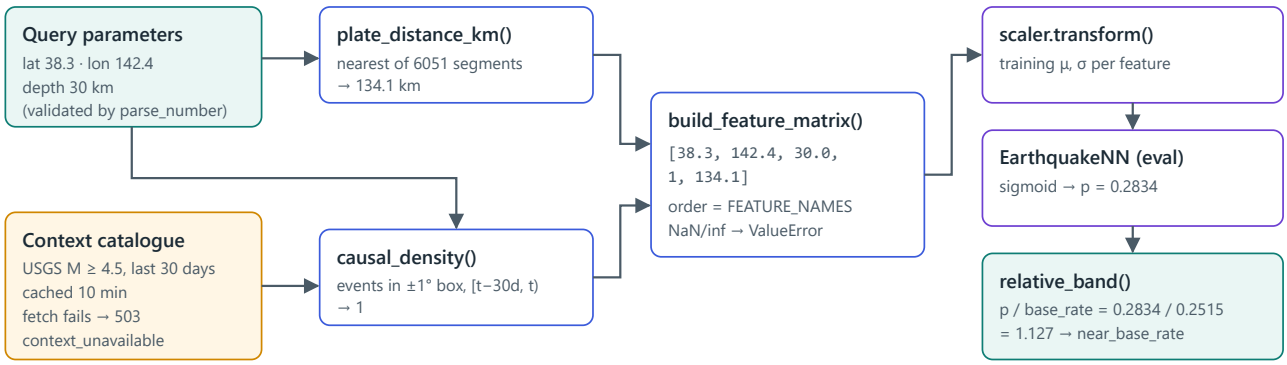
All numbers are from `model_meta.json` (trained 2026-10-18T21:35:45Z) on the **test period**, which was never used for training or early stopping.

Model	ROC-AUC	Average precision	Brier	Log loss
MLP (served)	**0.5703**	**0.3312**	**0.1970**	**0.5837**
Logistic regression (same features)	0.5524	0.3228	0.1967	0.5819
5° regional rate (training history, smoothed)	0.5242	0.2946	0.1989	0.5882
Constant training base rate (0.2515)	0.5000	0.2716	0.1982	0.5859

- **Brier skill score** against the base rate: 1 − 0.1970 / 0.1982 = **+0.006**.
- **Threshold 0.152** (chosen to maximise F1 on *validation*) applied to test gives precision 0.278, recall 0.971, F1 0.432 and accuracy 0.305. Confusion matrix: TN 96, FP 1,583, FN 18, TP 608.
- **Always predicting "< M5"** would have 72.8 % accuracy. This is why accuracy is not reported as the headline metric.
- **Calibration** on test: the predicted bins 0.1–0.2, 0.2–0.3 and 0.3–0.4 had observed rates of about 0.23, 0.28 and 0.34 (see `docs/model_evaluation.md` for the exact table). The probabilities are roughly calibrated but compressed into a narrow range.

Interpretation. The MLP ranks events slightly better than chance and slightly better than logistic regression, but its probability forecasts are almost no better than the base rate: logistic regression even has a marginally lower Brier score and log loss. No confidence intervals were computed, so the small differences between models should not be over-read. The honest summary is: **little to no practical predictive value**, as seismology predicts.

7.13 Inference walkthrough



Values are from a real request on 2026-10-18 (GET /api/predict?lat=38.3&lon=142.4&depth=30). lat/lon/depth flow into the matrix directly.

Figure 12. A real request traced through every step.

For `GET /api/predict?lat=38.3&lon=142.4&depth=30` on 2026-10-18:

1. `parse_number` accepts lat 38.3, lon 142.4, depth 30.0.
2. `get_context_catalog()` returns the cached M4.5+ catalogue (fetched 21:37:35Z, 613 events in 30 days).

3. `causal_density` counts events within 37.3–39.3°N, 141.4–143.4°E before the fetch time: **1**.
4. `plate_distance_km` finds the nearest boundary segment: **134.1 km**.
5. `build_feature_matrix` produces `[[38.3, 142.4, 30.0, 1, 134.1]]` in `FEATURE_NAMES` order.
6. `scaler.transform` standardises it. For example, the latitude becomes (38.3 − 8.85) / 33.31 ≈ 0.88.
7. `EarthquakeNN` in eval mode outputs **0.2834**.
8. `relative_band(0.2834)` : 0.2834 / 0.2515 = **1.127**, which falls between 0.8 and 1.25, so `near_base_rate` .
9. The JSON includes the features, the interpretation sentence, the model's test AUC and the context freshness. The modal shows "28.3 %" with "Near base rate · 1.13× of 25.2 %".

The bands are deliberately relative, not "low/medium/high risk": the absolute numbers cluster around the base rate, and the word "risk" would imply a probability of occurrence.

7.14 Evaluation risks that remain

- **A single test period.** Results for one three-month window may not generalise. A rolling-origin evaluation (Chapter 15) would be stronger.
- **Temporal correlation across the split boundary.** Aftershock sequences that start in training and continue into validation make adjacent periods related. A gap ("embargo") between splits would reduce this.
- **Spatial memorisation.** Latitude and longitude are raw inputs, so the model can memorise regional behaviour. It is not validated on regions absent from training.
- **Label noise.** Magnitude types (mb vs Mww) are not equivalent, and magnitudes are revised after review.
- **Out-of-distribution queries.** Scoring an arbitrary map point evaluates the model where no event occurred. Points far from any seismicity are outside the training distribution.

8 Analytical modules: sequences, forecasts, anomalies, aftershocks

Chapter 7 described the original classifier. This chapter describes the four analytical modules added in the `ml-platform` work. They live in the `analytics/` package, are trained or evaluated offline by `ml_pipeline.py`, and are served by the routes in `app.py`. Every number quoted here comes from the generated files in `models/`, which the Model Performance view and `docs/ml_evaluation.md` read as well.

8.1 What each module answers

Module	Question	Kind of method	Kind of output
A. Sequence Explorer	Which events in the current feed group together in space and time?	Unsupervised (DBSCAN)	Statistical grouping of observed events
B. Activity Forecasts	How many M ≥ 4.5 (or 5.0) events will a 10° cell record in the next 7 days?	Supervised count regression	Model estimate with an interval
C. Anomaly Detection	Is a cell's recent count unusually high compared with its own history?	Hypothesis test with multiple-comparison control	Statistical finding about observed data
D. Aftershock Analysis	How did activity evolve after a chosen earthquake, and does an Omori–Utsu decay describe it?	Descriptive statistics + maximum-likelihood point-process fit	Statistics, plus a model only when it passes checks

The layer table of §1.1 therefore gains two rows: *statistical findings* (Modules A and C, deterministic given the data and parameters) and *count forecasts* (Module B, a model estimate whose quality was measured out of sample). None of them predicts an individual earthquake.

8.2 The shared data layer

One validation rule set. Every module turns events into a table with `analytics.catalog.events_to_frame`:

- Rows without a finite time, a latitude in [−90, 90] or a longitude in [−180, 180] are dropped and counted.
- Duplicate ids keep the copy with the latest `updated` time.
- A missing magnitude or depth stays `NaN`. Such a row is never counted towards a magnitude threshold.

The feed client already de-duplicates. Applying the same rules again means the analytical code does not depend on that.

The historical catalogue. `python ml_pipeline.py download` fetches M ≥ 4.5 from 2000-01-01 to the most recent Monday (2026-10-19 for the committed models), in 120-day chunks:

- A chunk that reaches the 20 000-event limit is split in half until it fits.
- A failed chunk is retried and then aborts the download. The file therefore never contains a silent gap that would later look like zero activity.

The committed run downloaded 187 784 events in 82 chunks, with no invalid or duplicate records. The CSV is git-ignored; its JSON description records exactly what was requested.

The region grid (`analytics/regions.py`):

- 18 × 36 cells of 10° × 10°, with ids such as `R+30+140` (the lower-left corner).
- Latitude 90 belongs to the top row and longitude 180 wraps to −180.
- Neighbours wrap in longitude but not over the poles.
- Cells are equal-angle, so a cell at 60° covers half the area of an equatorial cell. Counts are only ever compared *within* a cell over time.

Weekly count matrices. `count_matrix` returns `C[r, w]`, the number of events with `mag ≥ threshold` in cell r and week w. Weeks are half-open, `[epoch + 7w d, epoch + 7(w+1) d)`, so an event exactly on a boundary belongs to the later week. A tiny tolerance makes "4.5 ≥ 4.5" true for decimal magnitudes.

Gaps versus zeros. A zero in a complete download is a real zero. A failed request is an error and never an empty table. In live data, a UTC day with no M ≥ 4.5 event anywhere on Earth is treated as a possible feed gap: the global rate is about 20 per day, and such a day never occurred in 2000–2026.

8.3 Module A — space-time sequence detection

Distance. Each event becomes an Earth-centred unit vector scaled by the chord length that corresponds to `eps_km`, plus a time coordinate scaled by `eps_hours`:

analytics/sequences.py

```python
def scaled_coordinates(lats, lons, times_ms, eps_km, eps_hours):
    """4-D coordinates in which the neighbourhood above is the Euclidean unit ball."""
    xyz = geo.unit_vectors(lats, lons) / geo.chord_for_arc_km(eps_km)
    t = (np.asarray(times_ms, np.int64) - int(np.min(times_ms))) / HOUR_MS / eps_hours
    return np.column_stack([xyz, t])
```

In these coordinates DBSCAN runs with `eps = 1`. Two events are neighbours when $(\text{chord}/\text{chord}(\varepsilon_km))^2 + (\Delta t/\varepsilon_h)^2 \le 1$:

- When Δt = 0, this is exactly "great-circle distance ≤ ε_km", because the chord grows monotonically with the arc.
- Points just either side of the antimeridian, or close to a pole, are close in 3-D, so neither case needs special code. This is the classic mistake avoided: Euclidean distance on raw degrees treats 179.9° and −179.9° as 360° apart, and a degree of longitude at 85° N as 111 km.

Why DBSCAN.

- It needs no number of clusters.
- It labels isolated events as noise, which is what most catalogue events are.
- It handles elongated fault-parallel groups.
- It is deterministic.

Alternatives were compared on the same data (see the evaluation below) and none replaced DBSCAN's defaults. Depth and magnitude are deliberately left out of the distance and only summarised afterwards.

What a sequence reports.

- **A stable id:** a hash of the earliest member's USGS id, so the same sequence keeps its id while that event stays in the window.
- **Counts:** members and core events.
- **Time:** start, end and duration.
- **Magnitude and depth:** distributions and histograms.
- **Location:** a spherical centroid, and an extent with an antimeridian-aware longitude span.
- **Linking statistics:** median nearest-member distance and the time gaps between events.
- **Ambiguous border events:** non-core events within reach of core events from two sequences (DBSCAN assigns them to just one).

The API also returns a plain-language explanation of the linking rule and a limitations list.

Evaluation (synthetic data with known labels, five seeds; never presented as real events):

Scenario	ARI (this method)	ARI, Euclidean on degrees	ARI, space only
dense clusters + uniform noise	1.000	0.998	1.000
two clusters at the same time, 400 km apart	1.000	1.000	1.000
same place, 20 days apart	1.000	1.000	0.000
cluster on the antimeridian	1.000	0.814	1.000
clusters at 84.5° and 89.9° N	1.000	0.474	1.000
duplicates (10) and invalid rows (4)	1.000	1.000	1.000
two overlapping sequences 60 km apart	0.000	0.000	0.000

The last row is a real limitation. At the default ε = 50 km the two overlapping sequences chain into one. They separate only near ε = 15 km (ARI 0.82), and an even smaller ε starts fragmenting them (10 km → 0.58).

Why they merge, and what was tried. DBSCAN joins two groups whenever a chain of core events links them (density connectivity). Two sequences whose epicentres spread 15 km around centres 60 km apart, starting 30 h apart, overlap enough for such a chain. Seven methods were run on twelve scenarios (the seven above plus overlaps at 30, 100 and 150 km, one 120 h apart, and a dense/sparse pair), ten seeds each:

Method	Mean ARI	Largest drop vs default	Notes
DBSCAN 50 km / 72 h (default)	0.817	—	merges the 60 km overlap
DBSCAN 25 km / 72 h	0.844	0.011	separates 100 km; 60 km in 2 of 10 seeds
DBSCAN 50 km / 24 h	0.824	0.008	separates the 120 h case only
DBSCAN 25 km / 36 h	0.846	0.028	
DBSCAN 15 km / 72 h	0.778	0.182	separates 60 km, leaves 5–10 % of members unassigned
HDBSCAN (min cluster 5 / 8)	0.785 / 0.766	0.538 / 0.709	separates 60 km, but absorbs or drops noise

The rule was fixed before the run: replace the default only if a method raises the mean ARI and no scenario falls by more than 0.02. Two methods passed (25 km / 72 h and 50 km / 24 h). The synthetic clusters are compact, though, and real aftershock zones of large earthquakes are long. So both were also run on the 99 real M7.5+ mainshocks with at least 20 events in their rupture zone within 30 days. This check was added after seeing the synthetic result, and is reported as such. With the default, 8 % of mainshocks had less than half of their zone's events in their own sequence. With 25 km it was 25 %, and with 24 h it was 17 %. Both split real sequences more often, so the defaults stay.

A flag-only check was also built: HDBSCAN inside each detected sequence, marking "possibly merged" when two or more subgroups each hold 20 % of the members. On synthetic data it flagged all 42 merged sequences and 27 of 396 single ones (7 %). But the first live run flagged the single M7.7 Panama sequence. Across the catalogue it fired on 53 of 177 M7+ mainshock sequences (30 %). Most of those are one aftershock sequence with structure along the rupture, so the flag would mostly be wrong. It is evaluated and documented, but not served. At 30 km and 30 h apart, no method separated the two sequences: locations and times alone cannot always say which sequence an event belongs to.

ARI and NMI are only meaningful because the labels are known. The real catalogue has no ground truth, so no accuracy is claimed for real sequences. Internal indices such as the silhouette score are not reported, because they assume convex clusters and have no notion of noise.

8.4 Module B — seven-day regional count forecasting

Target. For an origin T, a cell r and a threshold m:

> y(r, T) = number of USGS events with magnitude ≥ m and epicentre in r during [T, T + 7 d).

Features at the origin (`analytics/regional_features.py`). All of them are read from week columns *before* T:

Feature	Unit	Definition
`lag1`, `lag2`	events	Counts in the week before T and the week before that
`mean_4w`, `mean_13w`, `mean_52w`	events/week	Trailing mean weekly counts
`std_13w`	events/week	Population SD of the 13 weekly counts before T
`neighbor_mean_4w`	events/week	4-week mean summed over the up-to-8 neighbouring cells
`large_4w`	events	M ≥ 6 events in the cell in the 4 weeks before T
`weeks_since_last`	weeks	Weeks since the last week with a qualifying event (52 = none in a year)

The core of the implementation is a cumulative sum along the week axis, so any window sum is one subtraction:

`analytics/regional_features.py`

```
    def window_sum(cs, w):
        return cs[:, ks] - cs[:, ks - w]                # (regions, origins)
...
    # Index of the latest non-empty column strictly before each origin.
    cols = np.arange(Cr.shape[1])
    last = np.where(Cr > 0, cols[None, :], -1)
    last = np.maximum.accumulate(last, axis=1)
    last_before = last[:, ks - 1]
```

`cs[:, k]` is the sum of the columns *before* k, so the window ends exactly at the origin.

Leakage tests.

- One test changes the target week and every later week, then checks that the features do not change.
- Another builds a 70-week training matrix and a 52-week live matrix from the same raw events, and checks that the features at the same origin are bit-for-bit identical. Training and serving therefore cannot drift apart.

Temporal design.

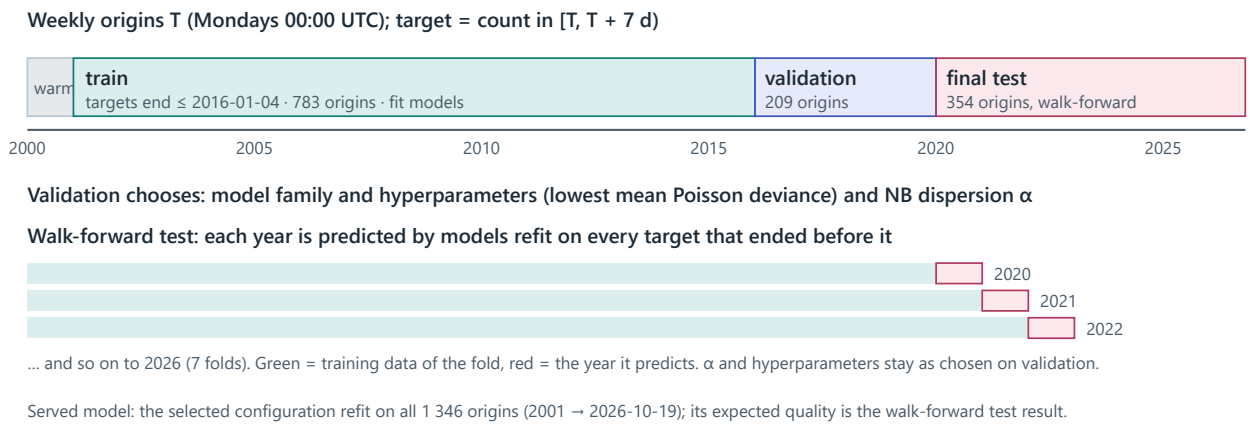


Figure 8a. Weekly origins from 2001 (after a 52-week warm-up). Validation chooses the model and the dispersion; the test period is evaluated walk-forward with yearly refits.

- **Train:** targets ending on or before 2016-01-04 (783 weekly origins).
- **Validation:** 2016-01-04 → 2020-01-06 (209 origins). It is used to choose the model, hyperparameters and NB dispersion.
- **Final test:** 2020-01-06 → 2026-10-19 (354 origins). For each calendar year, every model is refit on all samples whose target ends before the first origin of that year:

`analytics/forecast_training.py`

```
    for year in years:
        fold_k = np.array([k for k in k_test if datetime.fromtimestamp((epoch_ms + k * WEEK_MS) / 1000,
                                                                       tz=timezone.utc).year == year])
        k_fit = k_all[k_all + 1 <= fold_k.min()]
```

`k + 1 ≤ first test origin` means the target week of every training sample ends before the test year starts. A test asserts this ordering for every fold. Region eligibility (a 2001–2015 mean of at least 0.1 events per week: 131 cells at M4.5, 68 at M5.0) uses only the training period. The GLM's `log1p` and standardisation run inside its scikit-learn pipeline, so they are refit on each training window.

Candidates and selection.

- Three baselines: the historical mean of the training window, and the rolling 4-week and 52-week means.
- A Poisson GLM with four L2 strengths.
- Poisson-loss gradient-boosted trees in two sizes.

The model with the lowest validation mean Poisson deviance wins within each family. Trees are preferred over the GLM only if they are more than 1 % better, because the GLM is easier to explain. That rule picked trees at M4.5 (2.3 % better on validation) and the GLM at M5.0 (0.75 %). Deep learning was not tried: with nine features and about 100 000 sparse counts, it would add opacity without a plausible gain.

Uncertainty. Predictions are means. The predictive distribution is negative binomial (NB2), $\mathrm{Var} = \mu + \alpha(\mu)\mu^2$, with a dispersion that depends on the predicted mean, $\alpha(\mu) = \exp(a + b \cdot \log \mu)$. It is fitted by maximum likelihood on the *validation* predictions:

analytics/forecast.py

```
def fit_dispersion_curve(y, mu):
    """
    MLE of the dispersion curve alpha(mu) = exp(a + b log mu) given observed counts and model means.
    Starts from the constant-alpha MLE (b = 0), so the fit can only match or improve its likelihood.
    """
```

Low-mean cells turned out to be relatively more overdispersed ($b \approx -0.6$ at both thresholds). The curve was chosen over one constant α on validation data only: each half of the validation period fitted the dispersion and the other half scored it. The NB log score fell from 1.0598 to 1.0386 at M4.5, and from 0.7226 to 0.7188 at M5.0.

Intervals. The earlier intervals were equal-tailed NB quantiles, $[q_{0.1}, q_{0.9}]$ for "80 %". Most cells put more than 10 % of their predictive probability on zero, so the lower quantile is 0 and nothing is cut from the lower tail: the "80 %" interval actually held 93–97 % of the probability. That, rather than a wrong α, is why it covered about 95 % of test weeks. The served intervals are now the shortest integer intervals with at least the nominal probability:

analytics/forecast.py

```
def interval(mu, dispersion, level):
    """
    Shortest integer interval [lo, hi] whose predictive probability is >= level (ties: larger probability).
    Every lower bound from 0 to the (1 - level) quantile is tried with its smallest valid upper bound.
    """
```

No integer interval holds exactly 80 %. The evaluation therefore reports, next to the empirical coverage, the *model-implied* coverage: the probability the forecast gives its own interval. A calibrated forecast has the two close together, whatever the nominal level.

Results on the final test period (M4.5, 46 374 cell-weeks):

Model	MAE	RMSE	Poisson deviance	NB log score	80 % coverage
Gradient boosting (selected)	0.933	4.644	1.985	1.065	0.889
Poisson GLM	0.963	4.609	2.027	1.079	0.888
Historical mean	0.966	4.787	2.304	1.100	0.876
Rolling 52-week mean	0.971	4.765	2.175	1.090	0.883
Rolling 4-week mean	1.003	5.019	4.030	1.917	0.850

The means, and so MAE, RMSE and deviance, are exactly as before: only the predictive distribution and the intervals changed. How to read this honestly:

- Deviance falls by 13.9 % relative to the historical mean. Resampling whole test weeks gives a 95 % interval of 10.3–17.3 %. The model is better in every test year.
- MAE barely moves, because most cell-weeks are 0 or 1.
- The model beats the historical mean in 67 of the 131 cells and the 52-week mean in 94. The gain grows with activity: deviance skill is +3 % in the least active third of cells and +19 % in the most active, where ongoing sequences make recent counts informative.
- At M5.0 the selected GLM improves deviance by 6.6 % (2.9–9.9 %) but has a slightly *worse* MAE than the historical mean (0.553 vs 0.546). It beats the historical mean in only 24 of 68 cells.
- The rolling 4-week mean scores badly on deviance because it predicts exactly zero after a quiet month. Deviance and log score punish that harshly when an event then occurs. MAE does not.

Interval calibration on the test period (selected models):

Threshold	Construction	Empirical 80 % coverage	Model-implied	Mean width (events)
M4.5	equal-tailed, one α (previous)	0.958	0.942	2.86
M4.5	shortest, α(μ) (served)	0.889	0.863	1.61
M5.0	equal-tailed, one α (previous)	0.966	0.950	1.51
M5.0	shortest, α(μ) (served)	0.889	0.869	0.60

The intervals are 44 % and 60 % narrower, and the test NB log score improved at both thresholds (1.091 → 1.065, 0.740 → 0.736). They still cover about 2 points more than they claim, and the randomised PIT has too few weeks in its top decile (0.077 instead of 0.100): the predictive upper tail is still a little too heavy. Coverage also varies by cell, because the dispersion depends on μ but not on the cell.

Features that were tried and not added. An Omori-weighted (1/lag) sum of the last 52 weekly counts, the largest week in 52, and the 52-week neighbour mean were evaluated on validation only. The rule, fixed before the run: add them only if the selected family's validation deviance improves by more than 1 % in both halves of the validation period. They improved it by 0.67 % (M4.5) and 0.68 % (M5.0), so the feature set is unchanged.

Percentage errors (MAPE) are not reported, because they are undefined when the observed count is zero.



Figure 8b. Activity Forecasts on 2026-10-19 (headless Chrome, live data). Cells are coloured by the expected M4.5+ count for 2026-10-19 → 2026-10-26; the list gives each cell's 80 % interval and last week's observed count. The top cell is the M7.7 Panama sequence.

Serving. `/api/forecast` uses the cached 400-day history and fixes the origin at the start of the current UTC day. It builds the same features with `live_matrices` and runs the loaded estimator, which is never refit per request. It returns the mean, the 80 % and 95 % intervals, P(≥ 1), the baselines and the data window. `/api/forecast/<cell>` adds the 52 observed weeks and the walk-forward test forecasts for that cell, which the chart draws next to each other.

8.5 Module C — elevated observed activity

The operational definition is precise enough to be tested. Weekly counts in a cell are modelled as independent negative-binomial (NB2) counts with an unknown mean μ and dispersion α. The p-value is conditional on the total of the reference and observation windows, which removes μ from the test:

analytics/anomaly.py

```
def conditional_p_value(observed, reference_total, k, n, alpha):
...
    total = x + y
    out = np.ones(y.shape)
    pois, nb = (y > 0) & (a <= 0), (y > 0) & (a > 0)
    out[pois] = stats.binom.sf(y[pois] - 1, total[pois], k / (n + k))
    r = 1.0 / a[nb]
    out[nb] = stats.betabinom.sf(y[nb] - 1, total[nb], k * r, n * r)
```

The method:

- For each cell, the reference window (26 or 52 weeks) gives a weekly mean $\hat{\mu}$ and a total X. The observation window (k = 1, 2 or 4 weeks) gives a total y.
- The weekly dispersion α is the larger of the reference window's method-of-moments estimate and the cell's 2001–2015 **maximum-likelihood** estimate (`models/anomaly_reference.json`).
- With independent NB2(μ, α) weeks, the reference total is NB with shape n/α and the window total is NB with shape k/α, and both have the same scale. Given their sum N = X + y, the window total is BetaBinomial(N, k/α, n/α), whatever μ is. With

α = 0 this becomes Binomial(N, k/(n + k)), the classical conditional test for two Poisson counts.

- The one-sided p-value is P(Y ≥ y | N).
- Benjamini–Hochberg turns the p-values of every tested cell into q-values.
- A cell is **elevated** when q ≤ FDR and at least 3 events were observed.
- Cells with fewer than 5 reference events are `insufficient_reference`, never "normal".
- When feed gaps are suspected, unflagged cells are `data_incomplete`.
- `expected` = $k\hat{\mu}$ and `reference_sd` = $\sqrt{(k \cdot w + k^2 \cdot w/n)}$, with $w = \hat{\mu} + \alpha\hat{\mu}^2$, are shown for context. They do not enter the p-value. The response fields are the same as before.

How the test got here. Three versions were evaluated on the same null simulation: stationary NB weekly counts with each cell's *2016–2019* mean and variance (M4.5, 140 cells, 4-week window, 52-week reference). The long-term dispersion comes from 2001–2015, so it is never evaluated on the data it was estimated from. Every flag in this simulation is false. With every cell null, BH at FDR 0.05 should leave at most about 5 % of runs with a flag.

1. **Reference window only.** 35 % of runs raised at least one false alert.
2. **Plug-in NB with a moment floor.** The NB had mean $k\hat{\mu}$ and variance $k \cdot w + k^2 \cdot w/n$, and α had a floor at the 2001–2015 method-of-moments dispersion. 12 % of the first 200 runs had a false alert (13.65 % over 2 000 runs). Detection of a +4× excess after BH was 9 %.
3. **Conditional test with an MLE floor** (the current version).

Two diagnostic runs located the remaining problem in version 2. Given the simulation's *true* 2016–2019 dispersion but the estimated mean, 11.3 % of runs still had a false alert. Given both, 3.55 %. The excess therefore came mostly from the estimated mean, not from the dispersion. For heavy-tailed counts, 52 weeks usually *underestimate* μ, because the bursts that carry the mean are rare. The variance term $k^2 \cdot w/n$ is symmetric and cannot correct that skew. Conditioning removes μ instead of estimating it.

Power had a separate cause: the moment floor. One megathrust sequence dominates a 15-year variance but not a likelihood. Over the 140 simulated cells the moment estimate has quartiles 2.0 / 4.4 / 10.8, and the MLE has 1.0 / 2.4 / 4.6. With α around 4, a +4× excess almost never reaches the $p \approx 3 \times 10^{-4}$ that BH needs across 140 cells.

Comparison. All six variants saw the same 2 000 null runs and the same injection trials. The choice used only the null simulation and the 2016–2019 injection; the held-out columns were computed after it.

Variant	Null runs with a false alert	Detected +2× / +4×, 2016–2019	Detected +2× / +4×, held-out	Held-out flags (88 runs)
Previous: plug-in NB, moment floor	0.1365	0.0157 / 0.0973	0.0156 / 0.0913	70
Plug-in NB, MLE floor	0.1900	0.0470 / 0.1843	0.0435 / 0.1809	104
Plug-in NB, empirical null (99th percentile)	0.1265	0.0120 / 0.0476	0.0122 / 0.0492	58
Plug-in NB, empirical null (95th percentile)	0.2235	0.0463 / 0.2230	0.0448 / 0.2230	128
Conditional, moment floor	0.0345	0.0108 / 0.0671	0.0117 / 0.0626	42
Conditional, MLE floor (adopted)	**0.0560**	**0.0351 / 0.1549**	**0.0334 / 0.1506**	**80**
Diagnostic: true α, estimated mean	0.1130	—	—	—
Diagnostic: true α and mean	0.0355	—	—	—

The empirical null sets each cell's floor to the smallest α whose 4-week NB total reaches the cell's 2001–2015 99th (or 95th) percentile 4-week count with probability at least 1 % (or 5 %).

Reading the table:

- **Only the adopted variant improves both measures.** Runs with a false alert fall from 0.1365 to 0.056 (standard error 0.005). On the first 200 runs, which are identical to the earlier 200-run evaluation, the rate falls from 0.12 to 0.045. Held-out detection at +4× rises from 0.091 to 0.151 (12 592 trials).

- **Each change alone fails.** The conditional test with the moment floor is calibrated (0.0345) but loses power (0.063 at +4×). The MLE floor with the plug-in mean gains power but raises false alerts to 0.19. Together they work, because conditioning pays for the lighter floor.
- **The empirical null moves along the trade-off.** Depending on the quantile it trades false alerts for power, and nothing principled fixes the quantile.
- **Declustering was not evaluated.** The null simulation and the injection operate on counts, and declustering would remove the aftershock sequences that this module mainly finds. It remains future work, for a separate "background-rate change" question.

Remaining miscalibration. The adopted test is still slightly above nominal at the default setting, and more so with a 26-week reference. Each row below uses the same simulation design with 1 000 runs:

Threshold	Window	Reference	Cells	Previous	Conditional, moment floor	Adopted
M4.5	1 w	26 w	106	0.103	0.033	0.073
M4.5	1 w	52 w	140	0.076	0.028	0.044
M4.5	2 w	26 w	106	0.134	0.040	0.076
M4.5	2 w	52 w	140	0.111	0.039	0.065
M4.5	4 w	26 w	106	0.198	0.042	0.081
M4.5	4 w	52 w	140	0.152	0.037	0.063
M5.0	1 w	26 w	38	0.063	0.024	0.046
M5.0	1 w	52 w	65	0.056	0.019	0.040
M5.0	2 w	26 w	38	0.068	0.023	0.047
M5.0	2 w	52 w	65	0.063	0.024	0.044
M5.0	4 w	26 w	38	0.094	0.028	0.065
M5.0	4 w	52 w	65	0.080	0.028	0.051

- The adopted test is below the previous method in all 12 settings. It is at or below 0.05 in 5 of them, and its worst case is 0.081 (M4.5, 4-week window, 26-week reference).
- The conditional test with the moment floor stays below 0.05 everywhere, at the cost of power.
- The remaining false alerts come from cells whose 2016–2019 dispersion is several times anything in their 2001–2015 record. The cells with the most false alerts were near Vanuatu (true α 2.8 against a 2001–2015 MLE of 0.8), the South Sandwich Islands (36.6 against 9.2) and Kaikōura (19.2 against 0.5). No per-cell history can anticipate a regime that has not happened there yet.
- With 1 000 runs per setting the Monte Carlo uncertainty is about ±1.5 points. Four settings, all at M4.5, have their whole 95 % interval above 0.05 (worst 0.081, 0.066–0.100).

Why false alerts remain: three nulls. A separate random stream (2 000 runs each) ran the adopted test, the moment-floor test, and an oracle that knows each cell's simulated dispersion:

Null	Adopted	Moment floor	Oracle (true α)
NB with 2016–2019 mean and variance (as above)	0.076	0.049	0.018
Same, dispersion capped at each cell's 2001–2015 floor	0.017	0.004	0.016
Real 2016–2019 weeks resampled independently	0.398	0.231	0.065

- **The test is calibrated when its dispersion is right.** With the cap, and with the oracle, false alerts stay below 2 %. In 64 of the 140 cells the 2016–2019 dispersion exceeds the 2001–2015 floor; the excess comes from them.
- **The headline 5.6 % depended on the random stream.** On this independent stream the same null gives 7.65 %, so about 6–7 % is the honest summary.

- **Real counts are burstier than any NB.** Resampling real weeks puts a sequence week into the observation window at random, and the detector flags it in 40 % of runs. Even the oracle exceeds 5 %, because a single extreme week is rarer under a variance-matched NB than in reality. Under this construction a “false alert” is a real burst placed at random, which is exactly what the detector is for. The FDR level is therefore a property of the model, not a real-world false-alarm rate, and the API’s method text now says so.
- **What would remove the excess, and why it was not done.** A floor high enough to cover the 2016–2019 dispersion of every cell (the moment floor comes close) brings false alerts down to 3.5–4.9 % but cuts held-out detection at +4× from 15 % to 6 %. Neither variant improves both, so the MLE floor stays. A discrete (Tarone-type) multiplicity filter, which drops cells that could never reach significance, was checked too: no tested cell was in that position, so it would change nothing.
- A numerical fix came out of this work. SciPy’s beta-binomial upper tail is computed as 1 − CDF, which rounds to exactly 0 for very strong excesses. Those p-values are now summed directly from the probability mass function, so `p_value` is never reported as 0. No flag changed.

Power, measured by injection. Extra events were added to the last week of every unflagged tested cell, one cell at a time, in each of the 88 held-out 4-week windows (12 592 trials per factor). The earlier evaluation used one random cell per window (88 trials), so these estimates are about 12 times more precise:

Injected excess	Detected after BH	p ≤ 0.05 before correction	Rolling rule
+0.5 × expected	0.0014	0.0559	0.1832
+1 × expected	0.0039	0.1105	0.3154
+2 × expected	0.0334	0.2809	0.7272
+4 × expected	0.1506	0.6047	0.9085

The detector is still conservative. It flags strong excesses, which in practice means active aftershock sequences, and misses moderate ones. Most cells’ weekly counts are genuinely heavy-tailed, so a +4× excess is often within their normal range.

The overall rate hides a sparse-count effect. At +4×, detection is 0.4 % in cells expecting fewer than one event per 4-week window and 2.4 % at 1–2 events. Together these are 63 % of the trials, and +4× there means only a few extra events. Detection is 16 % at 2–5 expected events, 57 % at 5–10 and 63 % at 10 or more. No calibrated test can separate three or four extra events from chance among about 140 cells. The rolling rule catches more, but at about 200 times the false-alert rate (0.102 per cell test, against 0.00046).

Held-out behaviour, 2020–2026. There is no ground truth on real data. More flags is what higher power predicts; it is not evidence of better detection.

Threshold / window	Runs	Flags (adopted / previous)	Runs with a flag (adopted / previous)
M4.5, 1 week	354	99 / 77	0.2203 / 0.1695
M4.5, 2 weeks	177	93 / 74	0.3672 / 0.3051
M4.5, 4 weeks	88	80 / 70	0.5341 / 0.4659
M5.0, 1 week	354	48 / 38	0.1130 / 0.0904
M5.0, 2 weeks	177	34 / 32	0.1638 / 0.1469
M5.0, 4 weeks	88	33 / 30	0.2500 / 0.2273

Missing data. Removing 3 random days from 60 real windows was always detected by the gap check. It also removed 9.9 % of the flags, which is why those windows are reported as incomplete rather than normal.

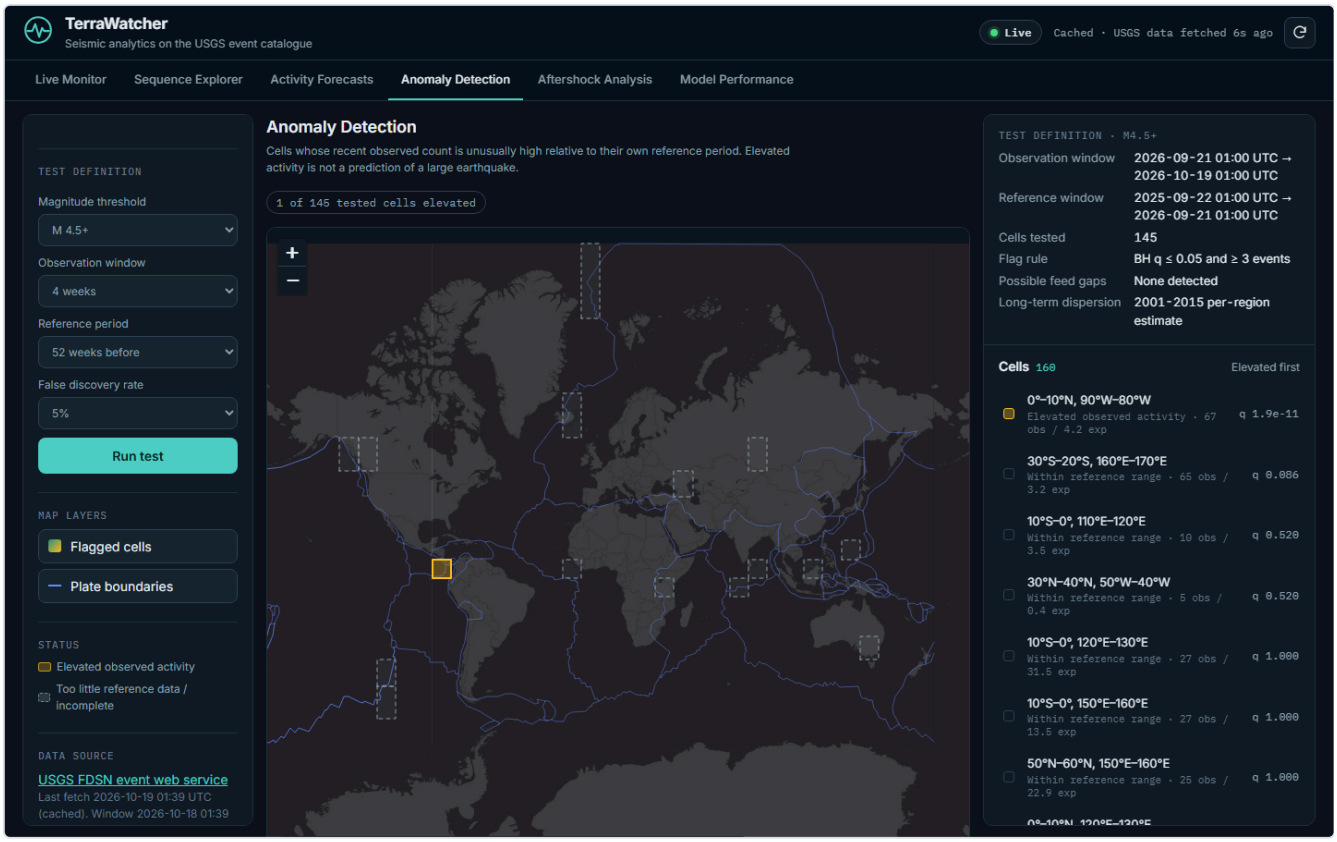


Figure 8c. Anomaly Detection on 2026-10-19, captured with the previous test (plug-in mean, moment floor). One cell is elevated (amber); dashed cells had too little reference data to be assessed. The inspector lists the test definition, then cells with their status and q-value.

Isolation Forest was considered and rejected. On sparse, overdispersed counts it has no explicit reference, and its scores cannot be read as probabilities.

Live, on 2026-10-19, the previous test flagged one of 145 tested cells: 0–10° N, 90–80° W, with 67 events against about 4.2 expected ($q = 1.9 \times 10^{-11}$). That is the M7.7 Panama sequence, and the UI says that is what it is. The adopted test was run on the downloaded catalogue for the same 4-week window (to 2026-10-19 00:00 UTC, M4.5). It flags two cells:

- the Panama cell ($q = 2.4 \times 10^{-10}$);
- 30–20° S, 160–170° E: 64 events against about 3.2 expected ($q = 0.0026$). This is the sequence of the M 6.7 Loyalty Islands earthquake of 2026-09-25. With its 2001–2015 moment floor ($\alpha = 18.6$) the previous test gave $q = 0.093$; the MLE floor is 4.5.

8.6 Module D — event-centred aftershock analysis

Selection (`analytics/aftershock.select`). It is deterministic and listed in every response:

- the great-circle distance to the reference epicentre is at most the radius (default: Wells & Coppersmith rupture length 10^(−2.44 + 0.59 M) km, clipped to 30–1000 km);
- the magnitude is at least the threshold, and events without a magnitude are excluded;
- optional depth limits apply, and once a limit is set, events without a depth are excluded;
- there is a pre-event window and a post-event window, and the post window never extends past the present;
- the reference event itself is excluded.

The radius query is done by USGS itself (`maxradiuskm`), so only nearby events are downloaded.

Statistics. The analysis returns:

- daily bins, hourly bins for the first three days, and logarithmic time bins (rates per day);
- the cumulative count;
- magnitude, depth and distance histograms;

- a conditional binomial test that the post-event rate exceeds the pre-event rate.

Daily bins hide the rapid early decay, and the log-time bins show it. A zero bin means no catalogue event met the criteria. Regional network outages cannot be detected from this query, and the response says so.

Model. λ(t) = μ + K/(t + c)^p, with the background μ fixed to the pre-event rate. The negative log-likelihood of an inhomogeneous Poisson process on [a, b] is:

analytics/aftershock.py

```
def _nll(theta, t, mu, a, b):
    logK, logc, p = theta
    K, c = math.exp(logK), math.exp(logc)
    lam = mu + K * (t + c) ** (-p)
    if np.any(lam <= 0) or not np.all(np.isfinite(lam)):
        return 1e12
    return -(np.log(lam).sum() - mu * (b - a) - _integral(K, c, p, a, b))
```

It is minimised with L-BFGS-B from nine starting points, on log K and log c so that both stay positive. The fit starts `fit_start_hours` after the event (6 h by default), because catalogues miss many early aftershocks. If the optimum puts c at its lower bound, c is not resolvable from data that start hours later. It is then fixed, K and p are re-estimated, and the UI explains this: it is a statement about resolution, not a failure.

When the fit is shown. It needs at least 20 events, and all of these checks must pass:

- the optimiser converged;
- no parameter is at a bound;
- the Hessian is positive definite, so standard errors and 95 % intervals exist;
- the KS test of the transformed times (which are uniform if the model is right) gives p ≥ 0.05;
- the AIC beats a constant-rate model by at least 2.

Otherwise the status is `unreliable` or `insufficient_data`, the reasons are listed, and no curve is drawn. When it is drawn, the curve covers only the fit window and is never extrapolated.

Evaluation across sequences. The analysis covered 356 M ≥ 7 mainshocks from 2001 to 2026. 25 others were excluded because a larger event occurred shortly before or within the window. With M4.5+ data, the 90-day fits were:

- reliable for 103 sequences;
- unreliable for 43;
- not attempted for 210, which had too few events.

The median p of the reliable fits was 1.32 (IQR 1.09–1.62), consistent with published values around 1. For an out-of-sample check, the model was fitted on days 0.25–30 and used to predict the count in days 30–90. The median |log((obs+1)/(pred+1))| was:

- 0.41 for Omori–Utsu;
- 0.94 for the last-week rate held constant;
- 1.25 for the background rate alone.

The Poisson 95 % intervals covered only 58 % of the outcomes. The prediction itself is not biased (median signed log ratio +0.02; the outcome was above it in 51.5 % of sequences), so the intervals were too narrow. Two separate causes were measured:

- **Parameter uncertainty.** A fit's parameters are uncertain, and extrapolating from day 30 to day 90 amplifies uncertainty in p. Drawing (log K, log c, p) from the fit's covariance (the inverse Hessian, now returned by the API) raises 80 %/95 % coverage from 0.48/0.58 to 0.67/0.82.
- **Variability beyond Poisson.** Aftershocks trigger their own aftershocks, and the model is only an approximation. Both make real counts vary more than a Poisson process allows, and the relative error does not shrink as sequences grow. With parameter uncertainty alone, sequences with 150 or more fitted events were covered only 41 %/55 %.

A negative-binomial count stage (Var = λ + dλ²) addresses the second cause. Its dispersion d had to be calibrated on *other* sequences, so it was fitted by maximum likelihood on the 56 mainshocks before 2013 (d = 0.19) and applied unchanged to the 47 later ones:

analytics/aftershock.py

```
    if dispersion > 0:
        lam = lam * rng.gamma(1.0 / dispersion, dispersion, len(lam))
    counts = rng.poisson(lam)
```

Mainshocks 2013–2026 (out of sample)	80 % coverage (95 % CI)	95 % coverage (95 % CI)	Median 95 % width
Poisson, parameters fixed (previous)	0.51 (0.37–0.65)	0.62 (0.47–0.74)	12 events
+ parameter uncertainty	0.68 (0.54–0.80)	0.83 (0.70–0.91)	26 events
+ NB count stage	**0.81 (0.68–0.90)**	**0.91 (0.80–0.97)**	34 events

The calibrated intervals are honest but wide: the upper end of a 95 % interval is typically 12 times the lower end. They were validated only for M ≥ 7 mainshocks, M4.5+ events, the rupture-length radius and a 30 → 90-day horizon. The application lets users choose other magnitudes, radii and windows, so it still shows only the fitted decay of past data and parameter confidence intervals, never a forecast of future counts. The API labels the parameter intervals as such (`confidence_note`).

The live example from §8.5 was the M7.7 Panama event `us6000u18k` with 67 events at M ≥ 4.5 in 9.3 days. Its fit gave p = 1.21 (95 % CI 0.92–1.49), a KS p-value of 0.68 and ΔAIC = 64.7 against the constant-rate model.

8.7 Lifecycle, artifacts and failure states

Step	Command	Writes
Download catalogue	`python ml_pipeline.py download`	`data/history_catalog.csv` + `.json` (git-ignored)
Train and evaluate forecasts	`python ml_pipeline.py forecast`	`models/forecast_m4.5.joblib/.json` , `forecast_m5.0.*` , `forecast_backtest.json`
Evaluate anomaly detection	`python ml_pipeline.py anomaly`	`models/anomaly_reference.json` (served), `anomaly_eval.json`
Synthetic clustering benchmark	`python ml_pipeline.py sequences`	`models/sequence_eval.json`
Omori–Utsu across sequences	`python ml_pipeline.py aftershock`	`models/aftershock_eval.json`
Everything + report	`python ml_pipeline.py all`	all of the above + `docs/ml_evaluation.md`

- **Determinism.** Seeds are fixed (`SEED = 42` for the trees, `default_rng(7)` for the simulations). Tree early stopping is off. Two consecutive `all` runs gave identical JSON apart from timestamps.
- **Remaining variation.** The catalogue itself changes, because USGS revises events.
- **Validated loading.** `ForecastModel` checks the artifact version, the feature-name list, the scikit-learn version, the region list and α. Any mismatch leaves the model "unavailable" with the reason. A wrong-version pickle is never used.
- **Degradation.** A missing forecast model gives `503 model_unavailable` and makes `/api/health` `degraded` . A missing `anomaly_reference.json` makes anomaly detection fall back to the window dispersion and say so. Missing evaluation files make the Model Performance view print "not evaluated yet". The live monitor never depends on any of these.

8.8 How the analytical views are built

The frontend stays framework-free. Three new files sit next to `core.js` and `dashboard.js` :

- `static/charts.js` returns SVG strings for line, step, bar, dot, band and error-bar charts:

- every axis is labelled with a unit;
- every chart has `role="img"` and a `<title>`;
- empty data produces an explicit message instead of an empty frame;
- log axes skip non-positive values.
- `static/analytics.js` holds the pure view logic:
- validators for every analytical response (a malformed payload is refused with a message);
- `createTracker()` for superseded requests;
- colour scales, the sequence-layer spec and the texts that explain flags and fits.
- `static/ml.js` contains the five views as accessible tabs (arrow keys, `aria-selected`, a hash such as `#forecast`). Each view has its own controls, module map layer, inspector pane and panel area. `dashboard.js` exposes a small `window.TerraApp` interface (`fetchJson`, the map, hooks for map clicks, feed reloads and "analyse this event").

Hand-offs connect the modules:

- an event in Live Monitor, the largest event of a sequence, or the largest event in an elevated cell opens Aftershock Analysis with that event pre-filled;
- for a sequence, the radius is preset to cover its extent.

Up to four aftershock analyses can be overlaid. The comparison warns when they use different magnitude, depth or window settings.

8.9 Questions to expect

- **"Is this earthquake prediction?"** No. The forecast is a weekly count for a 1100 km cell, and its skill comes mostly from persistence of ongoing sequences. It cannot anticipate a sequence that has not started. The README, the UI and this guide all say so.
- **"How do you know there is no leakage?"** Features are computed only from columns before the origin, and two tests enforce it: changing the future leaves the features unchanged, and training and serving features are identical. The test period is evaluated walk-forward, with fold boundaries also asserted by a test. α and every hyperparameter come from validation only.
- **"Why is the anomaly detector so insensitive?"** Because aftershock clustering makes weekly counts heavy-tailed, so an honest reference is wide. A detector calibrated on one quiet year flags far too often. Conditioning on the window total and a per-cell maximum-likelihood dispersion floor lowered the false alerts and raised detection at +4× from 9 % to 15 % at the same time. Moderate excesses stay undetectable at a 5 % run-level error rate. Each trade-off was measured, not assumed (§8.5).
- **"What would you improve first?"**
- ETAS (space-time branching) modelling, which unifies sequences, forecasts and aftershocks.
- Declustering before the anomaly test.
- Per-cell or covariate-dependent dispersion.
- Magnitude-of-completeness estimation per region and time (see Chapter 15).

9 Scientific validity and evaluation

9.1 Why earthquake prediction is so hard

Earthquake *prediction* means specifying time, location and magnitude narrowly enough to act on. After more than a century of research, it is not possible. The main reasons:

- **The process is hidden.** Faults rupture kilometres underground under stresses that cannot be measured directly.
- **No reliable precursor is known.** Proposed precursors (foreshock patterns, radon, animal behaviour, electromagnetic signals) have not held up in systematic tests. Many large earthquakes have no identifiable foreshock.
- **Magnitudes follow a scale-free law.** The Gutenberg–Richter relation, $\log_{10} N(\geq M) = a - bM$ with $b \approx 1$, means each magnitude unit is about 10× rarer. A rupture that starts small can stop small or grow large, so magnitude is largely unpredictable from where and when an event starts. TerraWatcher's near-zero skill is a small, concrete illustration of this.
- **There is little data on rare events.** Big earthquakes on a given fault recur over centuries, so a few years of data contains few relevant examples.

What *is* scientifically established is **probabilistic forecasting**, for example long-term hazard maps and short-term aftershock forecasts (Omori–Utsu decay, ETAS models). These give probabilities over regions and time windows, with stated uncertainty, and are evaluated prospectively. TerraWatcher implements neither.

9.2 Retrospective pattern recognition vs prospective forecasting

- **Retrospective:** fit a model to past data and measure it on other past data. Even with a chronological split, the developer has seen the whole dataset and could, consciously or not, tune choices to it.
- **Prospective:** freeze the model and the evaluation protocol, *then* collect new data and score the forecasts. Testing centres such as CSEP (the Collaboratory for the Study of Earthquake Predictability) exist for this purpose.

TerraWatcher's evaluation is **retrospective**, with a chronological holdout. That is the minimum honest standard for a time-dependent problem, not proof of forecasting ability.

9.3 Why random splits mislead

With a random split, an aftershock in the test set has its mainshock and sibling aftershocks in the training set, at nearly the same place, depth and time. The model can "recognise" the sequence rather than learn anything general, so test scores are inflated. Features that summarise activity over a window make this worse if the window extends *after* the event (the original project's density feature did). A chronological split plus strictly causal features removes both effects.

9.4 Spatial leakage and geographic generalisation

Even with a chronological split, the test events occur in the same subduction zones as the training events. A model that memorises "the Kermadec trench produces many M5+ events" can score well on a chronological split and still fail on a region it has never seen. To measure *geographic* generalisation, hold out whole regions (spatial blocks) for testing. This has **not** been done for TerraWatcher.

9.5 Class imbalance and misleading accuracy

With 27 % positives in the test period, a model that always says "not M5+" is 72.8 % accurate and useless. TerraWatcher therefore reports:

- **The confusion matrix:** TP (correctly flagged M5+), FP (flagged but < M5), FN (missed M5+), TN.
- **Precision** = TP / (TP + FP): of the events flagged, how many were M5+ (0.278).
- **Recall** = TP / (TP + FN): of the M5+ events, how many were flagged (0.971).
- **F1** = 2PR / (P + R): their harmonic mean (0.432).
- **ROC-AUC:** the probability that a random positive is scored above a random negative (0.5 means chance).

- **Average precision:** the area under the precision–recall curve. Its chance level equals the positive rate (0.272 here).

The high recall and low precision come from a low threshold (0.152). Almost everything is flagged, which is easy and uninformative. The threshold-free metrics (AUC, Brier) are the fairer summary.

9.6 Baselines

A metric means little without a reference. TerraWatcher compares against:

1. **The constant base rate**: always predict 0.2515. This is the "no skill" reference for Brier and log loss.
2. **The regional historical rate**: the share of M5+ events in each 5° cell during training, shrunk towards the base rate with 20 pseudo-counts. This captures "some regions produce bigger events".
3. **Logistic regression on the same features**: tells you whether the neural network's extra complexity buys anything.

9.7 Probability calibration

A forecaster is **calibrated** if, among all cases where it says 30 %, about 30 % happen. The **Brier score** (mean squared error of probabilities) rewards both calibration and sharpness. The **Brier skill score**, $1 - \text{Brier}_{\text{model}} / \text{Brier}_{\text{baseline}}$, is 0 for no skill and 1 for perfect. TerraWatcher's +0.006 means its probabilities are essentially no more informative than the base rate.

9.8 How to run an honest out-of-sample evaluation

1. **Fix the question and the label** before looking at results.
2. **Make every feature causal**: computable at prediction time from past data only.
3. **Split by time.** Add an embargo gap if events are clustered.
4. **Fit all preprocessing on training data only.**
5. **Tune only on validation data.** Touch the test set once.
6. **Compare with simple baselines** using proper scoring rules (log loss, Brier) and ranking metrics.
7. **Report uncertainty**, for example with bootstrap confidence intervals or several rolling test windows.
8. **Report negative results.** TerraWatcher's README and dashboard show its weak skill openly.

9.9 What evidence would be required to claim forecasting ability

- A forecast defined for **future time windows and regions**, not conditional on an event having occurred.
- **Prospective** testing on data collected after the model was frozen, ideally by an independent party.
- **Statistically significant improvement** over established reference models (for example Poisson or ETAS) on proper scoring rules, across many windows and regions.
- Robustness to catalogue revisions and to magnitude-type differences.

None of this exists for TerraWatcher. Its model is a **prototype and teaching component**. Its value in a portfolio is that it was built and evaluated *correctly* and its limitations are stated plainly, not that it predicts earthquakes.

Category	In TerraWatcher?	Example
Observed earthquake data	**Yes**	USGS events on the map
Historical patterns	**Yes**	Activity clusters, histograms
Model-derived estimates	**Yes** (exploratory)	Conditional severity score
Independently validated forecasts	**No**	—

10 Configuration and deployment

10.1 Supported runtime

Component	Version established by the project	Notes
Python	3.12 (verified with 3.12.10)	The development venv and the clean-clone test both used 3.12
Flask	≥ 3.1, < 3.2	3.1.3 installed
PyTorch	≥ 2.13, < 2.15	2.14.1 installed; CPU only
scikit-learn	≥ 1.7, < 1.8	must match the version that pickled `scaler.pkl` (1.7.2)
NumPy / pandas	NumPy ≥ 2.0, < 2.3; pandas ≥ 2.2, < 2.4	
Node.js	18+ (verified with 24.18)	only for the frontend unit tests
Browser	Any modern browser; acceptance-tested in Chromium	other browsers not tested

The project uses a plain virtual environment. Container files were intentionally not added, at the project owner's request.

10.2 Clean setup on a new machine

```
git clone https://github.com/Rase240/Real-Time-Earthquake-Monitoring-Geospatial-Analytics.git
cd Real-Time-Earthquake-Monitoring-Geospatial-Analytics
python -m venv .venv
.\.venv\Scripts\Activate.ps1          # Windows PowerShell
source .venv/bin/activate              # Linux / macOS
pip install -r requirements-dev.txt    # runtime deps + pytest
python -m pytest -q                    # expect: 188 passed
python app.py                          # http://127.0.0.1:5000
```

This exact sequence (clone, new venv, install, test, start the server, request `/api/health`) was run on 2026-10-19 in a fresh directory for the earlier version of the project (91 tests at the time). The current suite of 188 tests was run in the project virtual environment; a fresh-clone run of this version has not been repeated. The analytical modules need no extra step: their models and evaluation files are committed in `models/`.

10.3 Environment variables

No variables are required. Optional ones:

Variable	Default	Purpose
`TERRAWATCHER_HOST`	`127.0.0.1`	Bind address. `0.0.0.0` exposes the dev server to the network.
`TERRAWATCHER_PORT`	`5000`	Listening port
`TERRAWATCHER_DEBUG`	`0`	Flask debug mode (interactive debugger; never on a shared network)
`TERRAWATCHER_LOG_LEVEL`	`INFO`	`DEBUG`, `INFO`, `WARNING`, …
`TERRAWATCHER_USGS_URL`	`https://earthquake.usgs.gov/fdsnws/event/1`	Upstream base URL (useful for failure testing)
`TERRAWATCHER_USGS_TIMEOUT`	`15`	Seconds per USGS request
`TERRAWATCHER_USGS_RETRIES`	`1`	Retries on timeout/429/5xx
`TERRAWATCHER_FEED_CACHE_TTL`	`60`	Feed cache seconds
`TERRAWATCHER_CONTEXT_CACHE_TTL`	`600`	Context-catalogue cache seconds
`TERRAWATCHER_STALE_MAX_AGE`	`1800`	Maximum age of a stale copy

Example (PowerShell): `$env:TERRAWATCHER_PORT = "5001"; python app.py`. Example (bash): `TERRAWATCHER_PORT=5001 python app.py`.

There are **no secrets**. USGS, Esri tiles and unpkg require no credentials, and the repository contains no keys. (The original code accepted a map-tile API key in the URL query string; that option was removed.)

10.4 External services and artifact requirements

- **Network access** at runtime to `earthquake.usgs.gov` (server side) and to `server.arcgisonline.com`, `unpkg.com`, `fonts.googleapis.com` and `fonts.gstatic.com` (browser side).
- **Artifacts** that must sit in the project root together: `earthquake_model_v1.pth`, `scaler.pkl` and `model_meta.json`. If any is missing or inconsistent, the app runs without the model.
- **`data/tectonic_plates.geojson`.** Without it, `/api/health` reports `degraded`, `/api/plates` returns 404, and the model cannot compute `plate_distance_km`.

10.5 Frontend and backend startup

There is no separate frontend server and no build step. `python app.py` serves the page at `/`, the static files under `/static/` and the API under `/api/`. The browser uses relative URLs (`/api/earthquakes`), so the same code works on any host and port.

10.6 Health checks, ports, logs

```
curl http://127.0.0.1:5000/api/health
```

returns `{"status": "ok", "model_loaded": true, "plate_segments": 6051, …}`. The health check deliberately does not call USGS; upstream health is visible through the feed responses instead.

Logs go to standard output in the format `timestamp LEVEL logger: message`, for example `terrawatcher.usgs: USGS HTTP 503, retrying in 1.0s` or `terrawatcher: Serving stale cache for ('feed', 1, 2.5) …`. Set `TERRAWATCHER_LOG_LEVEL=DEBUG` for more detail. Werkzeug logs every request line.

10.7 Stopping, restarting, retraining

- **Stop:** press `Ctrl+C` in the terminal running `python app.py`.
- **Restart** after code changes: stop and start again (debug mode is off, so there is no auto-reload). A restart clears all caches.

- **Retrain:** `python train_model.py` (downloads about 16k events and takes roughly a minute) or `python train_model.py --offline`. Then restart the server, because artifacts are loaded at start-up.

10.8 Production considerations (not implemented)

> FUTURE The Werkzeug server prints “do not use it in a production deployment”. A real deployment would run `app:app` under a WSGI server such as Gunicorn or Waitress behind a reverse proxy with HTTPS. With several worker processes, each would hold its own cache, which is acceptable at this scale. None of this has been set up or tested.

11 Testing and debugging

11.1 Test suites

Suite	Tests	Framework	External services
`tests/test_usgs_client.py`	23	pytest + `unittest.mock`	`requests.get` mocked
`tests/test_features.py`	13	pytest	none (reads the bundled GeoJSON)
`tests/test_model.py`	9	pytest	none (loads committed artifacts; copies in `tmp_path`)
`tests/test_training.py`	7	pytest	none (synthetic catalogues)
`tests/test_api.py`	39	pytest + Flask test client	USGS replaced by `FakeUSGS`
`tests/test_geo_regions.py`	12	pytest	none
`tests/test_forecast.py`	29	pytest	none (synthetic matrices; committed artifacts)
`tests/test_sequences.py`	19	pytest	none (synthetic events)
`tests/test_anomaly.py`	26	pytest	none
`tests/test_aftershock.py`	14	pytest	none (simulated Omori–Utsu sequences)
`tests/test_api_analytics.py`	26	pytest + Flask test client	USGS replaced by `FakeUSGS` (with per-query `handler`)
`tests/frontend/core.test.js`	10	`node:test`	none
`tests/frontend/analytics.test.js`	16	`node:test`	none

Parametrised tests count once per case (for example, 8 invalid-input cases for `/api/predict`). Run:

```
python -m pytest -q
python -m pytest tests/test_api.py -q -k predict     # a subset
node --test "tests/frontend/*.test.js"
```

11.2 How USGS is faked

tests/conftest.py

```python
class FakeUSGS:
    """Replacement for usgs_client._get that records calls and returns canned payloads."""

    def __init__(self):
        self.query_payload = collection()
        self.count_payload = {"count": 0}
        self.error = None
        self.handler = None      # optional callable(path, params) -> payload, for tests that need per-query answers
        self.calls = []

    def __call__(self, path, params, timeout=None, retries=None, session=None):
        self.calls.append((path, dict(params)))
        if self.error is not None:
            raise self.error
        if self.handler is not None:
            return copy.deepcopy(self.handler(path, dict(params)))
        return copy.deepcopy(self.query_payload if path == "query" else self.count_payload)
```

The `fake_usgs` fixture monkeypatches `usgs_client._get` with an instance and clears both caches before and after each test. Tests then set `query_payload` (what USGS "returns"), `handler` (different answers per query, e.g. the event lookup and the radius search of the aftershock module) or `error` (how it "fails"), and can inspect `calls` to check what would have been sent.

Everything above `_get`, including normalisation, caching, routing and serialisation, runs for real. Because the fake replaces the whole transport, no test can reach the internet by accident.

11.3 Important tests explained

Test	Behaviour checked	Why it matters	A failure would mean
`test_query_uses_full_utc_timestamps`	`starttime` / `endtime` include `T06:30:00`	Guards the date-truncation bug that hid the latest day	24 h view silently missing recent events
`test_invalid_coordinates_are_rejected` (7 cases)	Empty, short, null, out of range, text and NaN coordinates all give `None`	No fabricated (0, 0) points	Fake events on the map
`test_timeout_is_retried_then_reported_as_504`	2 attempts, then 504	Retry policy and status mapping	Hangs or wrong codes
`test_plate_distance_across_antimeridian`	179.9° vs −179.9° ≈ 0.2° × cos 52°	Correct geometry at ±180°	Pacific features wildly wrong
`test_causal_density_counts_only_the_past_window`	Old, future and self excluded	Leakage prevention	Inflated, unfair evaluation
`test_density_feature_does_not_see_the_future`	Appending future data changes no past feature	Strongest no-look-ahead guarantee	Temporal leakage
`test_feature_schema_mismatch_is_refused`	Old feature names leave the model unavailable	Training/serving contract	Silent garbage predictions
`test_single_row_matches_batch`	Batch and single-row predictions agree	BatchNorm in eval mode	Request-dependent outputs
`test_training_is_reproducible_with_a_seed`	Same seed gives identical weights and outputs	Reproducible results	Unrepeatable metrics
`test_earthquakes_stale_cache_is_labelled`	Upstream failure serves the old copy with `stale: true`	Honest freshness	Old data shown as live
`test_hotspots_without_model_have_null_scores`	`model_probability` is `null`	No placeholder scores	Fabricated 0.5 values
`test_predict_without_context_does_not_guess`	503 `context_unavailable`	No density-0 fallback	Silently wrong scores
`test_unexpected_exception_is_generic_500`	The exception text is not returned	Information hiding	Internal details leaked
`core.test.js: normalizeEvents tolerates missing optional fields`	`null` mag/depth/place; `javascript:` URL dropped	UI robustness and safety	"NaN km", XSS link

11.4 Acceptance testing performed

The following were exercised against the live USGS API and recorded on 2026-10-18/19. They are manual checks, not automated tests:

1. **Load.** Dashboard loaded with 32 events (24 h, M2.5+) and no console errors.
2. **Filter.** "7 days" showed 337 events, equal to the API `count`. The strongest event (M 6.9) matched the API maximum.
3. **Depth filter.** "Deep" showed 8 events, equal to the number of API events with depth ≥ 300 km.
4. **Selection.** The detail panel matched the API record field by field (id `us6000u2xs`).
5. **Point analysis.** The UI value 27.2 % equalled `probability: 0.2719` from `/api/predict`.
6. **Upstream failure.** A second instance with an unreachable USGS URL showed an error banner and no events.

7. **Recovery.** After a simulated browser-side failure, Retry restored “Live”.
8. **Request supersession.** Three rapid window changes ended on the last choice, with the spinner hidden.
9. **Layouts.** 1440×900, 1200×850, 768×1024 and 375×812: no horizontal scroll; the drawer works on mobile.

The analytical views were checked on 2026-10-19 against live data at 1440×900 and 375×812:

10. **Sequences.** 30 days of M2.5+ gave 60 sequences among 1 847 events. Selecting the largest one (86 events, max M 7.7, Panama) highlighted its members, zoomed the map and drew its charts.
11. **Forecasts.** 131 cells were shown. The Panama cell showed 11.31 expected (80 %: 1–26) with its baselines, features and backtest.
12. **Anomalies.** With the current (conditional) test, 2 of 145 tested cells were elevated: Panama (67 observed vs 4.15 expected, $q = 2.4 \times 10^{-10}$) and New Caledonia (66 vs 3.15, $q = 0.0021$). The Panama cell’s explanation, weekly chart and contributing events matched `/api/anomalies/R+00-090`.
13. **Hand-off.** “Analyse activity after the largest event” opened Aftershock Analysis for `us6000u18k`: 67 events, fitted $p = 1.21$ (95 % CI 0.92–1.49), KS $p = 0.68$, ΔAIC 64.7.
14. **Model Performance** matched `models/*.json`.
15. **Degraded mode.** A second instance with `TERRAWATCHER_MODEL_DIR` pointing to an empty folder showed “Forecast unavailable … No forecast was computed” with Retry, and “Not evaluated yet” for each evaluation. Anomaly detection fell back to the reference-window dispersion with a notice (it then flagged more cells). The live monitor was unaffected.
16. **Mobile.** At 375×812 there was no horizontal page scroll in any view; wide tables scroll inside their cards.

Steps 12 and 15 used the previous anomaly test (plug-in mean, moment floor). They were not repeated in the browser after the conditional test was adopted. §8.5 gives the adopted test’s result for the same window, computed on the downloaded catalogue.

Rechecked on 2026-10-19 after the reliability changes (in-app Chromium, desktop size, a separate instance on port 5057 with live USGS data): Model Performance showed every new block from the regenerated artifacts without console errors. `/api/forecast` returned 131 cells with the new intervals and the dispersion curve, and `/api/anomalies` tested 144 cells (2 elevated) and reported its `method_id`. Sequence Explorer found 60 sequences in 30 days of M2.5+. Steps 9, 15 and 16 (layouts, degraded mode, mobile) and the hand-offs were not repeated after these changes. The screenshots were taken before them.

Not verified: Firefox and Safari, sustained real rate limiting by USGS, long-running memory behaviour, and screen-reader output (only the ARIA attributes were reviewed).

11.5 Troubleshooting reference

Problem	Diagnosis	Fix
`ModuleNotFoundError: flask`	venv not active	Activate `.venv` ; `pip install -r requirements-dev.txt`
Tests fail with `InconsistentVersionWarning`	sklearn ≠ 1.7.x	Reinstall from `requirements.txt` or retrain
Model card "Unavailable"	`GET /api/model-info` → `load_error`	Restore all three artifacts from git or retrain
"Feature schema mismatch" in `load_error`	`features.FEATURE_NAMES` changed without retraining	Retrain, or revert the feature change
Banner "Earthquake feed failed: Could not reach USGS"	No outbound network / DNS	Check `curl https://earthquake.usgs.gov/fdsnws/event/1/version`
504 `upstream_timeout`	USGS slow	Retry; raise `TERRAWATCHER_USGS_TIMEOUT`
"newest 2000 of N" in metrics	Query larger than `MAX_EVENTS`	Expected; narrow the window or raise the magnitude
Map grey, list fine	Esri tiles or unpkg blocked	Check the browser network tab; corporate proxies often block CDNs
400 `out_of_range` for `lat`	Latitude outside ±90 (often a lat/lon swap)	Check argument order
Browser CORS error	Page opened from a different origin (e.g. `file://`)	Open via `http://127.0.0.1:5000` ; the API is same-origin by design
Port in use	Another process on 5000	`TERRAWATCHER_PORT=5001`
Stale banner persists	USGS down for > 60 s	Data older than 30 min is never served; then an error appears

11.6 Debugging workflow

1. **Reproduce at the API level first:** `curl "http://127.0.0.1:5000/api/earthquakes?days=1"` . If the JSON is right, the bug is in the browser; if not, it is in Flask or upstream.
2. **Check the server log** for `ERROR` lines (`log.exception` prints full tracebacks).
3. **In the browser**, open DevTools: the Console shows warnings such as rejected events; the Network tab shows each `/api/*` call and its JSON.
4. **Write a failing test** using `fake_usgs` with the problematic payload, fix the code, and keep the test.

12 Security, reliability and performance

12.1 What is implemented

Area	Protection	Where
Input validation	Type, finiteness and range checks; integer-only `days` / `limit`	`parse_number`
Output encoding	All dynamic HTML escaped; event URLs restricted to `https://earthquake.usgs.gov/`	`escapeHtml`, `normalizeEvents`
Browser hardening	CSP (no inline scripts, same-origin `connect-src`), `nosniff`, `Referrer-Policy`, `frame-ancestors 'none'`	`_security_headers`
Third-party code	Leaflet pinned to 1.9.4 with SRI hashes	`index.html`
CORS	None emitted (same-origin only)	by design; tested
Error disclosure	Generic 500; upstream errors expose only the exception type	handlers, `_get`
Safe defaults	Bind 127.0.0.1, debug off, no secrets	`config.py`
Model loading	`torch.load(weights_only=True)`, schema validation	`ModelBundle`
Upstream reliability	Timeouts, bounded retries honouring `Retry-After`, labelled stale fallback	`usgs_client`, `cached_fetch`
Resource bounds	≤ 2000 events per response; cache ≤ 32 entries; 30-day maximum window	`config.py`
Dependencies	`pip-audit --path .venv/Lib/site-packages` (2026-10-19): no known vulnerabilities in the project's dependencies; 6 advisories in the venv's installer pip 25.0.1 (fixed in pip ≥ 26.2)	requirements files

12.2 Performance characteristics (measured locally)

Operation	Observed time	Notes
Server start-up	about 2 s	mostly importing PyTorch
`/api/earthquakes?days=1`, cache miss	about 1.4 s	dominated by the USGS round trip
`/api/earthquakes?days=30&min_mag=0`, cache miss	about 3.6 s	2000 events plus the `count` call
Any cached feed request	< 10 ms	
`/api/predict`, first call	about 2 s	includes the context-catalogue download
`/api/predict`, context cached	a few ms	5-feature forward pass
`/api/forecast`, history not cached	about 6 s	downloading ~8 600 global M4.5+ events (400 days) dominates
`/api/forecast`, history cached	10–25 ms	52-week matrices, features for 131 cells, one model call
`/api/anomalies`, history cached, new parameters	about 0.1 s	648-cell test + BH; then ~3 ms from `analysis_cache`
`/api/sequences?days=30`, feed not cached	about 2.8 s	USGS feed download + DBSCAN on ~1 850 events; ~8 ms when cached
`/api/aftershocks`, not cached	about 2.7 s	event lookup + radius query + MLE fit; ~2 ms when cached

These timings were observed during development on one Windows laptop; they are not benchmarks. The original code ran ~300 single-row model inferences and a pure-Python plate-distance loop on every feed request (4.7 s for a 300-event response). The feed no longer runs the model at all, and plate distance is vectorised with NumPy.

Map rendering uses Leaflet's canvas renderer (`preferCanvas`), which handles 2000 circle markers smoothly. The list renders at most 200 rows. Automatic refresh pauses while the tab is hidden.

12.3 Known limitations and recommendations (not implemented)

> NOT IMPLEMENTED
>
> - **No inbound rate limiting.** A client could request every `min_mag` value in a loop. The cache bound limits memory, not USGS traffic. Recommendation: a small per-IP limiter, or restricting `min_mag` to the UI's preset values.
> - **Development server.** Use a WSGI server for any public deployment (§10.8).
> - **Duplicate concurrent fetches.** Two simultaneous cache misses for the same key both call USGS (acceptable at this scale).
> - **The pickle in `scaler.pkl`** is trusted input. Storing μ and σ in JSON would remove pickle entirely.
> - **CDN dependency.** If unpkg is down, the map is unavailable. Vendoring Leaflet into `static/` would remove that dependency.
> - **No HTTPS** locally; required for any real deployment.

13 Important code walkthroughs

The walkthroughs follow execution order: configuration, server start, data fetching, features and model, an API route, then the browser.

13.1 config.py

Purpose. A single place for every tunable value.

Key content. The `_env_float` and `_env_int` helpers; paths built from `BASE_DIR` (`os.path.dirname(os.path.abspath(__file__))`), so the app works regardless of the current working directory.

config.py

```
MODEL_FILE = os.path.join(BASE_DIR, "earthquake_model_v1.pth")
SCALER_FILE = os.path.join(BASE_DIR, "scaler.pkl")
MODEL_META_FILE = os.path.join(BASE_DIR, "model_meta.json")
TECTONIC_FILE = os.path.join(BASE_DIR, "data", "tectonic_plates.geojson")
```

- **Inputs/outputs:** environment variables in, module constants out.
- **Connections:** imported by `app.py`, `usgs_client.py`, `train_model.py` and the tests.
- **Common mistakes:**
- Relative paths like `"scaler.pkl"` break when the app is started from another directory. The original code had a case-sensitivity problem with the GeoJSON file name for the same kind of reason.
- Tests that monkeypatch a config value (for example `config.MAX_EVENTS`) must patch `config.X`, not a copy imported elsewhere with `from config import X`.

13.2 app.py: start-up and a route

Purpose. Wire everything together and expose HTTP.

app.py

```
def feed_params():
    days = parse_number("days", default=1, lo=1, hi=config.MAX_DAYS, cast=int)
    min_mag = round(parse_number("min_mag", default=2.5, lo=-1.0, hi=10.0), 1)
    return days, min_mag


def get_feed(days, min_mag):
    def fetch():
        end = utc_now()
        start = end - timedelta(days=days)
        events, stats = usgs_client.fetch_events(start, end, min_mag=min_mag, limit=config.MAX_EVENTS)
        truncated = len(events) >= config.MAX_EVENTS
        total = usgs_client.count_events(start, end, min_mag=min_mag) if truncated else len(events)
        return {"events": events, "count": len(events), "truncated": truncated, "total_available": total,
                "query": {"days": days, "min_mag": min_mag, "start": iso(start), "end": iso(end),
                          "limit": config.MAX_EVENTS},
                "fetched_at": iso(end), **stats}

    return cached_fetch(("feed", days, min_mag), config.FEED_CACHE_TTL, fetch)


@app.route("/api/earthquakes")
def earthquakes():
    payload, cache = get_feed(*feed_params())
    return jsonify({**payload, "source": SOURCE_NAME, "cache": cache})
```

Line by line:

- `feed_params` validates and normalises the inputs. Rounding `min_mag` to 0.1 bounds the number of distinct cache keys. The feed is split out of the route because `/api/sequences` clusters exactly the same cached feed: both endpoints share one download per `(days, min_mag)`.
- `fetch` is a **closure**: it captures `days` and `min_mag`, and `cached_fetch` calls it only on a miss. The time window is computed *inside* `fetch`, so a cached payload keeps the window it was actually fetched for.
- `truncated` is inferred from hitting the limit. Only then does the code pay for the extra `count` call.
- `**stats` merges `dropped_invalid` and `duplicates_removed` into the payload.
- The final line adds per-response metadata (`cache`) *outside* the cached object, so cached payloads are never mutated.

Edge cases:

- `days=1.5` gives 400 (cast to `int` fails).
- `min_mag=-1` is allowed, because USGS reports small negative magnitudes for tiny events.
- An exactly-2000-event result is reported as truncated, with `total_available` = 2000. That is conservative and correct.

13.3 usgs_client.fetch_events

`usgs_client.py`

```
def fetch_events(start, end, min_mag=None, limit=None, timeout=None, retries=None, session=None, circle=None):
    """Fetch and normalise events in [start, end). Newest first."""
    params = {
        "format": "geojson",
        "starttime": _iso(start),
        "endtime": _iso(end),
        "orderby": "time",
        "eventtype": "earthquake",
    }
```

- `orderby=time` returns newest first, so when `limit` truncates, the *oldest* events are dropped. That is what a "recent events" view needs.
- `timeout`, `retries` and `session` are injectable: the historical download uses `timeout=120, retries=4` for large chunks, and tests could pass a fake session.
- `circle=(lat, lon, radius_km)` adds the FDSN great-circle search (`latitude`, `longitude`, `maxradiuskm`) used by the aftershock module. USGS does the radius filter server-side, so only nearby events are downloaded.
- Passing `params=` to `requests` URL-encodes the values. Building URLs with f-strings (as the original code did) risks encoding bugs.
- **Common mistake:** formatting timestamps without a timezone conversion. `_iso()` calls `astimezone(timezone.utc)` first.

13.4 features.py and ml_model.py

These were covered in depth in §5.6, §7.5 and §7.11. The essential point for maintenance: `FEATURE_NAMES` **is the contract**. To change a feature you must:

1. Change the computation in `features.py`.
2. Update `FEATURE_NAMES`.
3. Retrain, which writes a new `model_meta.json`.
4. Restart the server.

If you skip step 3, the server refuses to load the old artifacts instead of serving wrong predictions.

13.5 dashboard.js: start-up and wiring

`static/dashboard.js`

```
document.addEventListener('DOMContentLoaded', () => {
  initMap();
  bind();
  loadPlates();
  refreshAll();
});
```

- The map is created first, so the layers exist before data arrives. `static/ml.js` registers its own `DOMContentLoaded` handler after this one (handlers run in registration order), so the map already exists when the analytical views add their module layer.
- `bind()` attaches every event listener exactly once and starts the two intervals: refresh every 3 minutes and the freshness label every 30 seconds.
- `loadPlates` and `refreshAll` (feed plus activity cells, plus the active analytical view through the `refresh` hook) run concurrently. Each handles its own errors, so one failure cannot block the rest.

Common mistakes:

- Calling `L.map('map')` twice ("Map container is already initialized"), prevented by the guard in `initMap`.
- Binding listeners inside a render function, which multiplies them on every refresh. Here, row listeners are attached to freshly created elements each render, and the old elements are discarded with `replaceChildren`.

13.6 train_model.main: guarding the evaluation

train_model.py

```
    train, val, test = chronological_split(df)
    assert train["time"].max() <= val["time"].min() <= val["time"].max() <= test["time"].min()

    X = {k: d[F.FEATURE_NAMES].to_numpy(float) for k, d in (("train", train), ("val", val), ("test", test))}
    y = {k: d["label"].to_numpy(int) for k, d in (("train", train), ("val", val), ("test", test))}
    scaler = StandardScaler().fit(X["train"])
    Xs = {k: scaler.transform(v) for k, v in X.items()}
    base_rate = float(y["train"].mean())
```

- The `assert` documents and enforces the chronological property. Note that `python -O` strips asserts; the property also holds by construction, because `chronological_split` sorts first.
- Columns are selected **by name** (`d[F.FEATURE_NAMES]`), never by position, which is what makes the feature order explicit.
- `scaler.fit` sees only training rows. `base_rate` also comes from training only, and it is the value shown to users as the reference.

14 Interview preparation

14.1 The 30-second introduction

“TerraWatcher is a real-time earthquake analytics dashboard I built with Flask and Leaflet. It pulls live data from the USGS earthquake API, validates and caches it, and shows events, statistics, recent activity clusters and plate boundaries on an interactive map. It also has a small PyTorch model, and the part I’m proudest of is how I evaluated it. I used a chronological split, made sure there was no future-data leakage, and compared it against baselines. It turned out to be barely better than the base rate, and the app says so openly instead of presenting it as a forecast.”

14.2 The two-minute technical explanation

“The backend is a single Flask app. A client module talks to the USGS FDSN service with timeouts and a retry policy for rate limits and server errors. It validates every record: anything without valid coordinates or a timestamp is dropped rather than guessed. Results go into a small thread-safe TTL cache. If USGS goes down, I serve the last good copy for up to 30 minutes, but it’s explicitly flagged as stale, and the UI shows a warning.

The frontend is plain JavaScript and Leaflet with canvas rendering. Pure logic like validation and statistics is in a separate module that I unit-test with Node. It cancels superseded requests when you change filters quickly, and every failure state is visible and recoverable.

For the ML part, the question is: given that an M4.5+ earthquake was recorded, is it M5 or bigger? The features are location, depth, distance to the nearest plate boundary, and the number of M4.5+ events nearby in the previous 30 days, computed strictly from the past. I trained on 24 months of data with a 70/15/15 chronological split. The MLP got a ROC-AUC of 0.57 on the newest three months, versus 0.55 for logistic regression and 0.50 for the base rate, with a Brier skill of essentially zero. That’s consistent with Gutenberg–Richter, so I present it as an exploratory component, not a predictor.”

14.3 The five-minute architecture walk-through

Use Figure 1 and cover, in order:

1. **One process, three concerns.** Serving the UI, proxying and normalising USGS data, and running the model. Explain why there is no database: all data is either fetched live or static.
2. **The data path** (Figure 2). Validation on both sides. The cache keys, TTLs and stale semantics. Truncation reporting with the `count` endpoint.
3. **Error contract.** One JSON error shape; status codes by cause (400 client, 502/504 upstream, 503 not ready, 500 bug). The UI shows `error` directly.
4. **The ML contract.** `features.py` shared by training and serving; `model_meta.json` stores the feature names; the loader refuses mismatches. Artifacts are loaded once at start-up and inference is batched.
5. **The evaluation story** (Figures 9–10). The leakage you fixed, the chronological split, the baselines, the honest result.
6. **Trade-offs.** No marker clustering (canvas is enough at 2000 points), in-process cache, dev server, and what you would change for production (Chapter 15).

14.4 Model, features, evaluation and leakage

Q1. What exactly does your model predict?

P(magnitude ≥ 5.0 | an earthquake of magnitude ≥ 4.5 has been recorded, with these five features). It classifies events that have already happened. It does not predict whether, when or where earthquakes occur. On the map, a score for an arbitrary point is the hypothetical “if an M4.5+ event happened here”.

Q2. Why did you choose M4.5 as the catalogue threshold?

Below about M4.5 the global catalogue’s completeness depends on the regional seismic network. An M2.5 catalogue would include many small Californian events and few from remote oceans, so the label would partly encode station density. I didn’t measure that effect directly; it is a design decision based on how the catalogue is assembled.

Q3. Walk me through the features and their units.

Latitude and longitude in degrees, depth in km, `local_density_30d` (a count of M4.5+ events within ±1° in the 30 days strictly before the event), and `plate_distance_km` (the distance to the nearest of 6051 plate-boundary segments, using a local equirectangular approximation with longitude wrapping).

Q4. How did you prevent data leakage?

Three ways. The density feature counts only events strictly before each event's time, and a test checks that appending future data changes no past feature. The split is chronological. The scaler and the base rate are fitted on the training split only. The original version had real leakage: its density counted the whole six-month window, including aftershocks that happened after the event.

Q5. Why not a random train/test split?

Earthquakes cluster in space and time. With a random split, an aftershock in the test set has its siblings in training, so the model recognises the sequence and the score is inflated. A chronological split mimics real use: train on the past, evaluate on the future.

Q6. What were your results, and how do you interpret them?

On the newest 15 % (2026-07-11 to 2026-10-18): ROC-AUC 0.570, average precision 0.331, Brier 0.197. Logistic regression got 0.552 AUC and 0.197 Brier; the constant base rate got 0.500 and 0.198. The Brier skill score is +0.006. It ranks slightly better than chance but has almost no practical value, which is what Gutenberg–Richter scaling predicts.

Q7. Why report Brier score and AUC instead of accuracy?

Positives are about 25 %. Always predicting "not M5" gives 72.8 % accuracy with zero usefulness. AUC measures ranking, and Brier and log loss measure probability quality. Since the app shows probabilities, calibration matters.

Q8. Why an MLP, if logistic regression does as well?

The MLP was inherited from the original project, and I kept it so I could evaluate it honestly. The comparison shows the extra capacity buys very little; logistic regression even has a slightly lower log loss. If I were starting fresh, I'd ship logistic regression for simplicity and interpretability. That's a recommendation, not something I changed.

Q9. How did you choose the decision threshold?

To maximise F1 on the validation period (0.152), then I applied it unchanged to test: precision 0.28, recall 0.97. The low threshold flags almost everything, which is why I treat the threshold metrics as secondary. The app itself shows probabilities relative to the base rate, not a binary label.

Q10. Why BatchNorm, and what can go wrong with it?

It stabilises training. The pitfall is inference: BatchNorm must use its running statistics, so the model has to be in `eval()` mode. In training mode, a single-row request can't be normalised. The loader calls `model.eval()`, and a test checks that single-row and batch predictions match.

14.5 API design, async requests, geospatial rendering and errors

Q11. Describe your API error contract.

Every error is `{error, code, status}`. 400 is for invalid input, with codes `missing_parameter`, `invalid_parameter` and `out_of_range`. 502/504 are for upstream failures. 503 means not ready (`model_unavailable`, `context_unavailable`). 500 is a generic message for bugs, with the traceback only in the server log. The frontend displays `error` directly, so messages are written for humans.

Q12. Why does the backend proxy USGS instead of the browser calling it directly?

It centralises validation, caching and rate-limit protection; it lets the model use the same data; and it gives one consistent schema to the frontend. It also keeps the browser same-origin, so no CORS is needed.

Q13. What happens if a user changes filters quickly?

Each `loadFeed` call aborts the previous `AbortController`. The aborted request throws a `superseded` error that is ignored, and only the current request may hide the loading indicator. I tested three rapid clicks: the UI ended on the last selection.

Q14. How do you handle the GeoJSON coordinate order?

GeoJSON is `[lon, lat, depth]` and Leaflet is `[lat, lon]`. I unpack the array exactly once, in `normalize_feature`, into named fields, and everything downstream uses names. There's a unit test for the mapping.

Q15. What is the antimeridian problem and where did you handle it?

Longitudes jump from +180 to −180, so naive differences across the Pacific are about 360° off. I wrap differences into [−180, 180) in both plate distance and density. That matters for Fiji, Tonga and the Aleutians, which are very active areas.

Q16. How does the map stay fast with up to 2000 events?

Leaflet's canvas renderer draws all markers on one canvas instead of creating 2000 SVG nodes. The list caps at 200 rows. The depth filter runs client-side without network calls. Refresh pauses when the tab is hidden.

Q17. How do you distinguish fresh, cached and stale data?

Every feed response has a `cache` object `{hit, age_seconds, stale, warning}` and a `fetched_at` timestamp. The top bar shows "Fresh", "Cached" or "Stale copy" with the age, and stale data also triggers an amber banner with the server's warning.

Q18. What happens when USGS is down?

The client retries once. If there's a cached copy under 30 minutes old, it's served with `stale: true`; otherwise the API returns 502/504. The UI shows a banner with Retry, keeps any previously loaded events, and never invents data. `/api/predict` returns 503 instead of scoring with a missing feature.

Q19. How did you validate input like `lat=nan`?

`float("nan")` parses successfully in Python, and NaN fails every comparison, so a range check alone would let it through. `parse_number` explicitly rejects non-finite values. There's a parametrised test with `lat=nan&lon=inf`.

14.6 Defects discovered and fixed during the engineering audit

Q20. What was the most surprising bug you found?

The "24 hours" view didn't show the last day. The USGS query used `endtime=YYYY-MM-DD`, which means midnight UTC today, so everything since midnight was excluded. Live testing showed the newest event was 23 hours old. I switched to full ISO timestamps and added a test.

Q21. What was wrong with the original hotspots?

They were seven hard-coded coordinates scored with a constant density of 5, and the endpoint returned a made-up 0.5 when the model was missing. I replaced them with observed activity clusters, the 2° cells with the most M4.5+ events in the last 30 days, and the model score is `null` when unavailable.

Q22. Were there data-integrity issues?

Yes. Missing coordinates were replaced with 0.0, which plotted fake events at 0°N 0°E, and missing depth became 10 km. Responses were silently capped at 300 events. Now invalid records are dropped and counted, optional fields stay null, and truncation is reported with the true total from the `count` endpoint.

Q23. What was wrong with the original ML evaluation?

Four things. A random split, accuracy as the only metric, no baselines, and a density feature that both leaked future events and was computed three different ways in training, the feed endpoint and the predict endpoint. Every observed event was also being scored, even though its magnitude was already known.

Q24. How did you make sure training and serving can't drift apart?

One shared `features.py`; `model_meta.json` stores the feature names; and the loader refuses artifacts whose schema differs, whose scaler has the wrong width, or whose weights don't fit the architecture. A pinned scikit-learn version plus a pytest warning-as-error catches pickle-version drift.

Q25. Did you find dependency issues?

`pip-audit` flagged 24 advisories in torch 2.5.1 and one each in requests and pytest. I upgraded the pins, retrained under the new versions (with identical metrics, because training is seeded), and verified in a fresh clone with a new venv. The audit is now clean.

14.7 Architecture trade-offs, reliability and scalability

Q26. Why no database?

There's nothing to persist. Events are fetched live from the authoritative source, and reference data is a static file. A database would add migrations and operations work without a feature that needs it. Historical analysis beyond 30 days would justify one (Chapter 15).

Q27. How would this scale to many users?

Most requests hit the 60-second cache, so USGS load stays roughly constant regardless of the number of users. For more traffic I'd run several WSGI workers. Each has its own cache, which is fine. If USGS traffic still mattered, a shared cache or a background refresher would be the next step. I haven't load-tested it.

Q28. Why an in-process cache instead of Redis?

One process, small payloads, and a cache that is fine to lose on restart. Redis would add a service to run for no measurable gain at this scale.

Q29. Is the cache thread-safe?

Yes, a lock guards the `OrderedDict`. The fetch runs outside the lock, so two simultaneous misses can both call USGS. That's a deliberate simplicity trade-off; a per-key lock would prevent it.

Q30. What's the weakest part of the system?

Scientifically, the model has almost no skill. Operationally, it runs on the development server, has no inbound rate limiting, and depends on CDNs for the map library and tiles.

14.8 Challenging the scientific validity

Q31. "So your app predicts earthquakes?"

No. It shows observed earthquakes and recent activity. The model only estimates, for an event that has already been recorded, whether it is M5+, and even that is barely better than the base rate. Exact earthquake prediction isn't scientifically possible today.

Q32. "Aren't the activity clusters effectively a forecast?"

They are counts of past events. Clustering does carry some short-term information (aftershocks), but quantifying that requires models like ETAS evaluated prospectively. I don't implement that, and the UI labels clusters "observed activity, not a forecast".

Q33. "AUC 0.57 — isn't that just noise?"

It's possible. I evaluated one three-month window without confidence intervals, so I can't claim significance. A rolling-origin evaluation with bootstrap intervals would answer that. The Brier skill of +0.006 already says the probabilities add almost nothing.

Q34. “What would convince you a forecasting model works?”

Forecasts for future windows and regions, frozen before the data arrives, evaluated prospectively on proper scoring rules, and consistently beating standard references like ETAS across many regions and periods, ideally verified independently, as in the CSEP testing centres.

15 Extension roadmap

Each item is a **proposal**, not an implemented feature. They are ordered by value relative to effort.

#	Proposal	Impact	Difficulty
1	Rolling-origin evaluation with bootstrap intervals	High	Low
2	Ship logistic regression (or keep both) and report calibration plots	Medium	Low
3	Store scaler parameters as JSON instead of pickle	Medium	Low
4	Embargo gap between splits; spatial-block validation	High	Medium
5	Historical explorer (> 30 days) backed by SQLite	High	Medium
6	Browser end-to-end tests (Playwright) in CI	Medium	Medium
7	WSGI server + inbound rate limiting for deployment	Medium	Medium
8	Vendor Leaflet; fallback tile provider	Medium	Low
9	Accessibility audit with a screen reader; data-table alternative to the map	Medium	Medium
10	Aftershock-rate baseline (Omori–Utsu / ETAS) as a scientifically grounded comparison	High	High

1. Rolling-origin evaluation. Retrain on months 1…k and test on month k+1, for many k, then report the mean and the bootstrap confidence interval of each metric. *Benefit:* it tells you whether 0.57 AUC is stable or noise. *Approach:* a loop around `chronological_split` and `train_nn` in a new `evaluate.py`. *Limitation:* more compute, and the windows are still retrospective.

2. Simpler model. *Benefit:* fewer parameters, interpretable coefficients and equal skill. *Approach:* `LogisticRegression` is already fitted in `train_model.py`, so save it and switch `ModelBundle`. *Limitation:* the same low skill; this is a simplification, not an improvement.

3. JSON scaler. *Benefit:* removes the pickle trust issue and the version coupling to scikit-learn. *Approach:* save `mean_` / `scale_` to `model_meta.json` and apply them with NumPy.

4. Embargo and spatial blocks. *Benefit:* quantifies leakage from aftershock sequences and tests geographic generalisation. *Approach:* drop 30 days around each split boundary; hold out continents or 10° blocks. *Limitation:* fewer samples per split.

5. Historical explorer. *Benefit:* real historical analysis (trends, b-value estimation per region). *Approach:* a periodic download into SQLite with indexes on time and location; new endpoints with pagination. *Limitation:* adds storage and a refresh job, the first real reason for a database.

6. Browser E2E tests. *Benefit:* automates the manual acceptance checks in §11.4. *Approach:* Playwright with route interception to fake `/api/*`. *Limitation:* adds a Node dependency and CI time.

7. Deployment hardening. A WSGI server, HTTPS via a reverse proxy, a per-IP limiter and restricted `min_mag` values.

8. Front-end resilience. Serve Leaflet from `static/` and configure a second tile source.

9. Accessibility. Test with NVDA or VoiceOver; offer a sortable table view as a non-visual alternative to the map; check colour contrast of the magnitude palette against WCAG AA.

10. Aftershock baseline. *Benefit:* a meaningful scientific reference for any future forecasting claim. *Limitation:* requires seismological expertise to implement and evaluate correctly. It would still be research-grade, not an operational forecast.

Infrastructure such as Kubernetes, message queues or microservices is deliberately **not** on this list: nothing in the current workload needs it.

16 Quick reference and glossary

16.1 Glossary

Term	Meaning
Antimeridian	The ±180° longitude line; longitude differences must be wrapped across it
Average precision	Area under the precision–recall curve; chance level = positive rate
Base rate	Share of positives in training data (0.2515); the no-skill forecast
BatchNorm	Layer that normalises activations; uses running statistics in `eval()` mode
Brier score	Mean squared error of probability forecasts (lower is better)
Brier skill score	$1 - \text{Brier}/\text{Brier}_{\text{baseline}}$; 0 = no skill
Causal feature	Computable at prediction time using only past data
Chronological split	Train on older data, test on newer data
Depth class	Shallow < 70 km, intermediate 70–300 km, deep > 300 km
Epicentre / hypocentre	Surface point above the rupture start / the rupture start itself (with depth)
FDSN	International Federation of Digital Seismograph Networks; USGS's web-service standard
GeoJSON	JSON format for geographic features; coordinates are `[lon, lat(, depth)]`
Gutenberg–Richter law	$\log_{10} N(\geq M) = a - bM$, $b \approx 1$: each magnitude unit is about 10× rarer
Leakage	Information unavailable at prediction time entering training or evaluation
Magnitude type	Method used to compute magnitude (mb, ml, Mww, …); not interchangeable
ROC-AUC	Probability a random positive outranks a random negative; 0.5 = chance
Stale data	A cached copy served because the upstream failed; always flagged
TTL	Time to live: how long a cache entry counts as fresh
Web Mercator	The map projection used by Leaflet tiles (EPSG:3857)

16.2 Key files

File	One-line role
`app.py`	Flask routes, validation, cache, errors, headers
`usgs_client.py`	Robust USGS access and normalisation
`features.py`	The five feature definitions (the training/serving contract)
`ml_model.py`	Network definition and validated artifact loading
`train_model.py`	Download → features → chronological split → train → evaluate → save
`config.py`	Every tunable, environment-overridable
`static/core.js`	Pure, tested browser logic
`static/dashboard.js`	UI state, networking, map and rendering
`model_meta.json`	Schema + metrics written by training, read by the server and the UI

16.3 API summary

Method and path	Key parameters	Returns
`GET /api/earthquakes`	`days` 1–30, `min_mag` −1–10	events + freshness + truncation
`GET /api/hotspots`	`limit` 1–50	observed 2° activity clusters
`GET /api/predict`	`lat`, `lon` / `lng`, `depth`	conditional severity score + features
`GET /api/model-info`	—	schema, training, metrics, baselines
`GET /api/health`	—	readiness, per module
`GET /api/plates`	—	plate-boundary GeoJSON
`GET /api/sequences`	feed params + `eps_km`, `eps_hours`, `min_samples`	detected sequences + assignments
`GET /api/forecast[/<cell>]`	`threshold` 4.5 / 5.0	7-day expected counts, intervals, baselines (+ history, features, backtest)
`GET /api/anomalies[/<cell>]`	`threshold`, `window_weeks`, `reference_weeks`, `fdr`	per-cell status, observed vs expected, p, q (+ weekly counts, events)
`GET /api/aftershocks`	`event_id`, `radius_km`, `pre_days`, `post_days`, `min_mag`, depths, `fit_start_hours`	statistics, Omori–Utsu fit or reasons, members
`GET /api/model-performance`	—	every module's evaluation results

16.4 Common commands

Task	Command
Install	`pip install -r requirements-dev.txt`
Run	`python app.py`
Python tests	`python -m pytest -q`
Frontend tests	`node --test "tests/frontend/*.test.js"`
Download the 2000–now catalogue	`python ml_pipeline.py download`
Train/evaluate all analytical modules	`python ml_pipeline.py all`
One module	`python ml_pipeline.py forecast` (or `anomaly`, `sequences`, `aftershock`, `report`)
Retrain the legacy classifier	`python train_model.py` (or `--offline`)
Lint (tool venv)	`ruff check .`
Health	`curl http://127.0.0.1:5000/api/health`
Build this guide	`pip install -r requirements-docs.txt` then `python docs/build_guide.py`
Check guide excerpts only	`python docs/build_guide.py --check`

16.5 Architecture in one paragraph

A single Flask process serves a static Leaflet dashboard and a JSON API. It fetches events from the USGS FDSN service with timeouts and retries, validates and de-duplicates them, and caches them for 60 s (10 min for the 30-day M4.5+ context catalogue), with labelled stale fallback for up to 30 minutes. The browser re-validates responses, filters by depth locally, and renders canvas markers, lists and statistics. A PyTorch MLP, loaded once at start-up with schema checks, scores hypothetical M4.5+ events using five causal features shared with the offline training script.

16.6 Model assumptions and limitations

- The score is conditional on an M4.5+ event occurring; it is not a probability of occurrence.
- Test ROC-AUC 0.570; Brier skill +0.006 against the base rate: little practical value.
- One retrospective chronological test period; no confidence intervals; no spatial holdout.
- Magnitude types are mixed; the catalogue is revised over time.
- Plate distance is approximate and the boundary data dates from 1987.

16.7 Demonstration checklist

- ☐ Run `python -m pytest -q` before the demo (188 passed) and start `python app.py`.
- ☐ Open the dashboard; point out the "Live / Fresh" status and the fetch time.
- ☐ Switch 24 h → 7 days → 30 days; show the count and the "newest 2000 of N" truncation note at M 0+.
- ☐ Apply the "Deep" filter; explain that it is client-side.
- ☐ Select a large event; open its USGS page from the detail panel.
- ☐ Toggle plate boundaries and activity clusters; open a cluster popup and read the "not a forecast" note.
- ☐ Click the map near a subduction zone; explain the conditional score and the base-rate comparison.
- ☐ Sequence Explorer at 30 days: select the largest sequence, read why its events were grouped, then hand it to Aftershock Analysis.
- ☐ Activity Forecasts: pick the busiest cell; compare the forecast with its baselines and show the out-of-sample backtest line.
- ☐ Anomaly Detection: open the flagged cell, read the explanation and the q-value, and say what "elevated" does not mean.
- ☐ Aftershock Analysis: show the log–log rate plot, the fit diagnostics, and an event whose fit is rejected (and why).
- ☐ Model Performance: state validation vs test, the baseline comparison and the cells where the model is worse, without being asked.
- ☐ Show `docs/ml_evaluation.md` and run `python ml_pipeline.py report` to show it is generated.
- ☐ If asked about failures: start a second instance with a bad `TERRAWATCHER_USGS_URL` and show the error state.
- ☐ Close with what you would do next (Chapter 15), labelled as future work.